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SECTION 02

TITLE: PROJECT REPORT

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1.0 Company Background

The Koperasi Kakitangan KADA Berhad is a cooperative organization established to serve the welfare and economic interests of the employees of the Kemubu Agricultural Development Authority (KADA) in Malaysia. Operating from its headquarters at Ibu Pejabat KADA Lundang in Kota Bharu, Kelantan, the cooperative plays a pivotal role in enhancing the financial well-being and professional growth of its members.

Vision and Ambitions:

Koperasi Kakitangan KADA Berhad aspires to be a leading cooperative, committed to uplifting the socio-economic status of its members through diversified financial services and investment opportunities. The cooperative aims to foster a culture of mutual assistance and economic empowerment, ensuring that all members benefit from collective growth and prosperity.

Achievements and Recognition:

Since its inception, the cooperative has been instrumental in providing financial solutions tailored to the needs of KADA employees. Through prudent management and member-centric initiatives, it has established itself as a reliable entity, contributing significantly to the financial stability and advancement of its members.

Collaborations and Initiatives:

Koperasi Kakitangan KADA Berhad collaborates closely with KADA and other financial institutions to offer a range of services, including savings schemes, loan facilities, and investment opportunities. These collaborations are designed to provide members with access to financial resources that support personal development and economic resilience.

Community Impact:

Beyond financial services, the cooperative is dedicated to fostering a sense of community among KADA employees. It organizes various programs and activities aimed at professional development, social engagement, and the overall well-being of its members. By promoting solidarity and mutual support, Koperasi Kakitangan KADA Berhad contributes to a cohesive and empowered workforce within the agricultural sector.

About Koperasi Kakitangan KADA Berhad

As an integral part of the KADA ecosystem, Koperasi Kakitangan KADA Berhad continues to evolve, adapting to the changing needs of its members and the broader economic landscape. Through its unwavering commitment to service and excellence, the cooperative remains a cornerstone of support for KADA employees, driving forward the mission of agricultural development and community enrichment in Malaysia.

2.0 Problem background

Currently Koperasi KADA has been using manual system management for the cooperative for 15 years. To address the issues from the current system, the issues will be broken down into a few points for a better understanding before proposing a new solution to the issues faced.

1. Manual Data Entry Process

As mentioned, Koperasi KADA requires staff to manually enter the application submitted by the applicant into the system. This process is highly inefficient as staff need to manually key in all the details for every application (membership application, loan application & membership termination). This will likely increase the human errors such as typos, incorrect data input and missing information as it can lead to inconsistencies in the records. Moreover, staff will spend some time repetitively keying in the data instead of focusing on higher-value activities. This issues will cause bottlenecks in operation

2. Inconvenient Process of Obtaining and Submitting Physical Form

Applicants who want to become a member of Koperasi KADA need to get a physical form directly in person at the office of Koperasi KADA and return back to the office to submit the form. As this manual process is inconvenient, it will discourage the potential applicant from joining and be part member of Koperasi KADA. Moreover, the form submitted might be misplaced during the submission and applicants need to submit the form again and need to go back and forth to the office in order to get the form and submit the form again. This issue is the same as a member of Koperasi KADA who wants to submit a loan application or membership termination application.

3. Lack of Self-Service Access to Personal Financial Statement

Members of Koperasi KADA need to go to the office in person to access their financial status (savings, loan balance, etc) as they can independently access their financial information. This is inconvenient for those who live far away from the office and for those that have busy schedules where it becomes a burden for them to care about their financial status. As they cannot access their financial status at their own convenience, it will reduce member satisfaction which might lead to membership termination.

4. Inefficient Reporting Mechanism

Due to its reliance on manual data management processes, it is unable to generate an accurate and timely report. These issues will lead to inconsistencies and delay in generating reports such as monthly reports of membership application, loan application, syer etc and even annual reports. The lack of accurate reporting makes it difficult for the cooperative to monitor the financial status or even track member activities. Besides, without timely and accurate report cooperative unable to make a plan strategically for future growth and demands

3.0 Proposed Solution

In this section, we will propose solutions that directly address the issues stated in the previous section

1. Automation of Data Entry Process

To eliminate inefficiencies and reduce human errors, the proposed system will feature automated data entry and validation. When applicants submit their membership, loan, or termination applications online, their details will be directly stored in the database, removing the need for manual keying by staff. The system will include built-in validation checks to ensure all required fields are completed correctly, reducing typos and missing information. This will streamline operations, minimize data inconsistencies, and allow staff to focus on higher-value tasks rather than repetitive data entry.

2. Online Application Submission

To enhance accessibility, applicants will be able to fill out and submit membership, loan, and termination applications online through a dedicated web-based system. This eliminates the need to visit the office in person, reducing inconvenience for both applicants and staff. Additionally, the system will store all application records securely, ensuring that no forms are lost or misplaced. Applicants will also receive real-time status updates via the system, keeping them informed about their application progress.

3. Self-Service Access to Financial Statements

A secure online member portal will allow Koperasi KADA members to view their financial statements, including savings balances, loan statuses, and transaction history. Members will be able to log in anytime, anywhere, eliminating the need for physical visits to the office. This self-service functionality will increase member satisfaction, improve financial transparency, and reduce the workload for staff handling financial inquiries.

4. Automated and Real-Time Reporting

The system will include automated report generation tools that compile real-time financial and membership data into structured reports. Monthly and annual reports for membership applications, loan applications, and syer contributions will be automatically generated and accessible to administrators. The system will also provide data analytics tools for tracking financial trends, member activities, and overall cooperative performance. This will improve decision-making, support strategic planning, and enhance operational efficiency.

4.0 Development Approach

For the development of the KADA Corporation System, we adopted the MVC (Model-View-Controller) architecture as the foundation for coding and structuring the system, along with the Agile development methodology to ensure flexibility and continuous improvement throughout the project. The MVC architecture allowed us to separate the system into three distinct layers: the Model layer, which handled database structures and managed critical data such as member information, transactions, and user roles; the View layer, which focused on designing an interactive and user-friendly interface for different user types, including members, ALK, and administrators; and the Controller layer, which served as the intermediary, processing user requests, executing business logic, and connecting the Model and View components. By following the Agile methodology, we worked in iterative development cycles, allowing us to make continuous enhancements, adapt to changing requirements, and ensure high-quality system performance. This approach enabled our team to collaborate efficiently, identify and resolve issues quickly, and implement new features seamlessly, making the KADA Corporation System well-structured, scalable, and adaptable to future improvements.

5.0 System Design (ERD, Use Case & Sequence Diagram)

5.1 Entities Relationship Diagram (ERD)

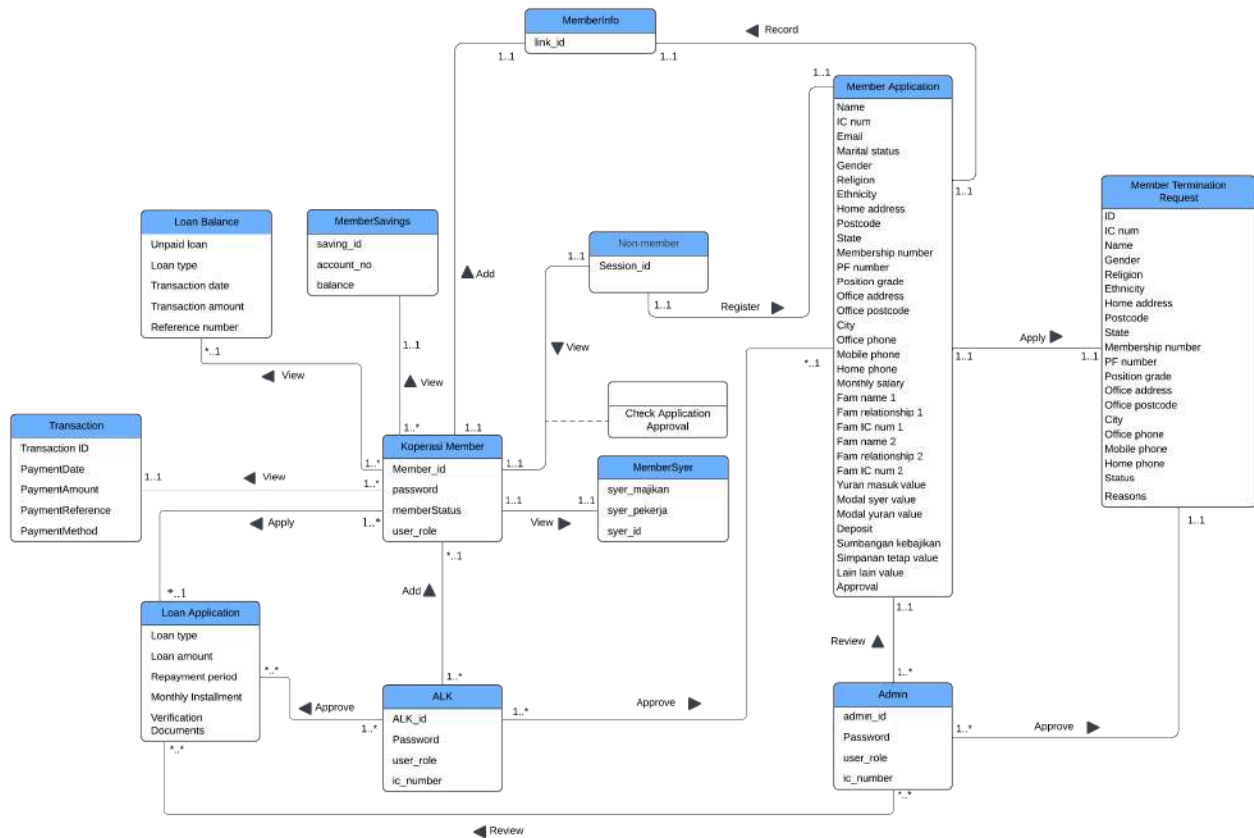


Figure 1: ERD Diagram of Koperasi KADA system

Reference Link:

<https://drive.google.com/file/d/1sRHFm270d1rm9ZB8H4piEvthHacxP6iC/view?usp=sharing>

5.2 Use Case Diagram

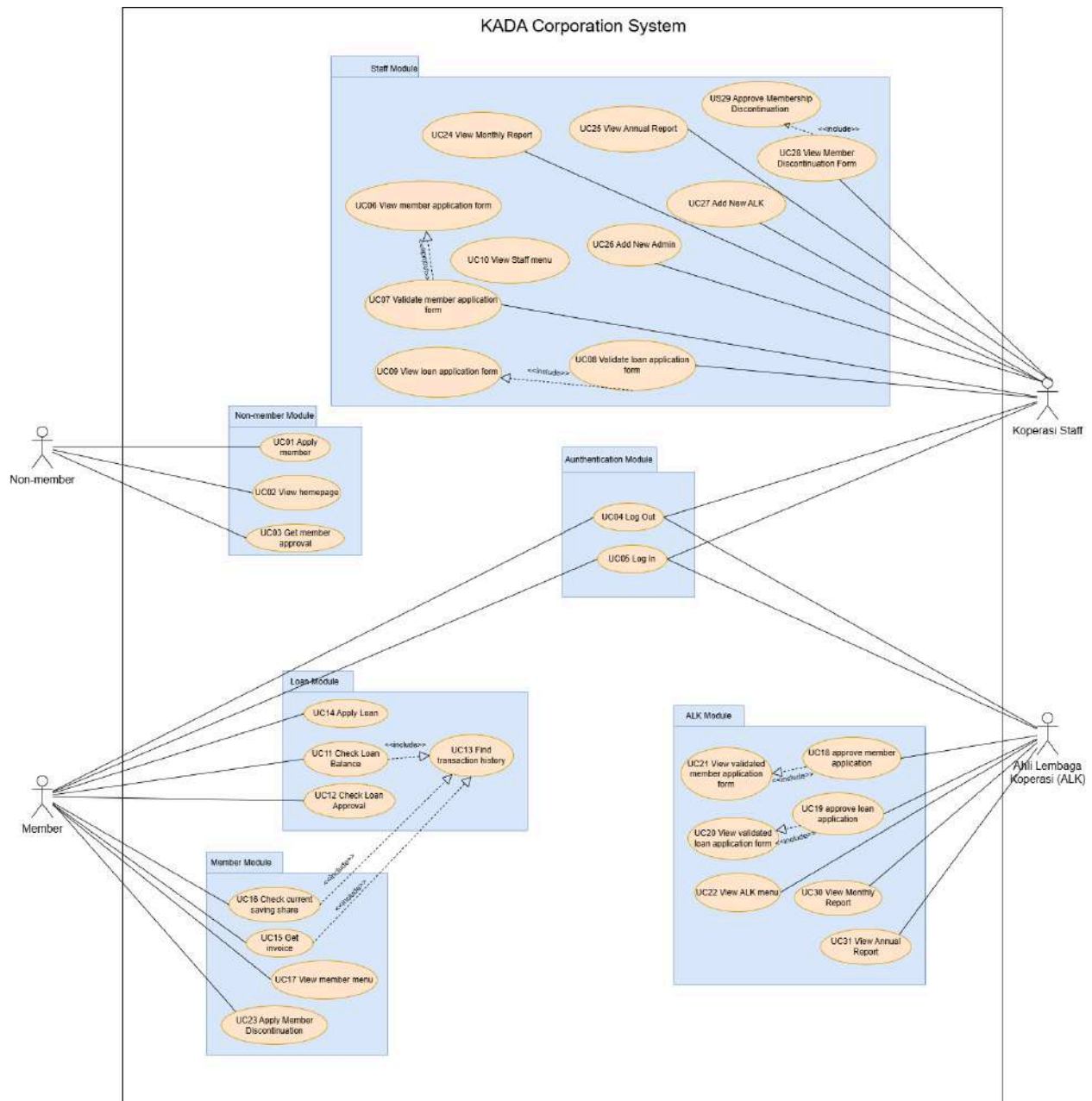


Figure 2: Use Case Diagram for Koperasi KADA system

5.3 Sequence Diagram

5.3.1 Module <Member>

5.3.1.1 UC17: Use Case View Member Menu

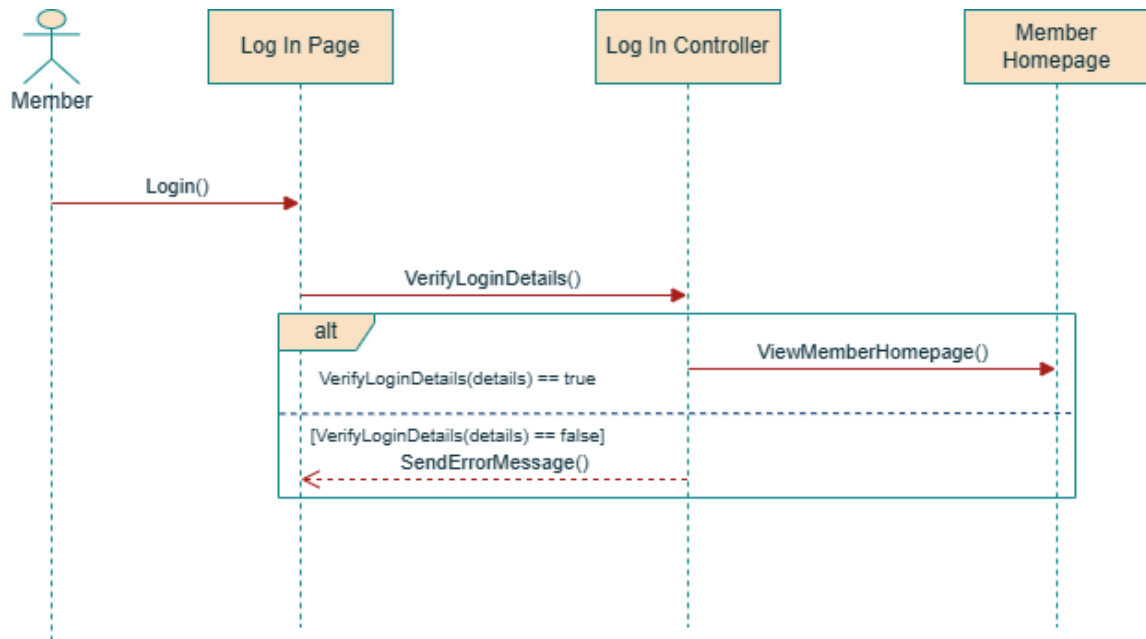


Figure 3: Sequence Diagram for View Member Menu

5.3.1.2 UC16: Use Case Check Current Saving

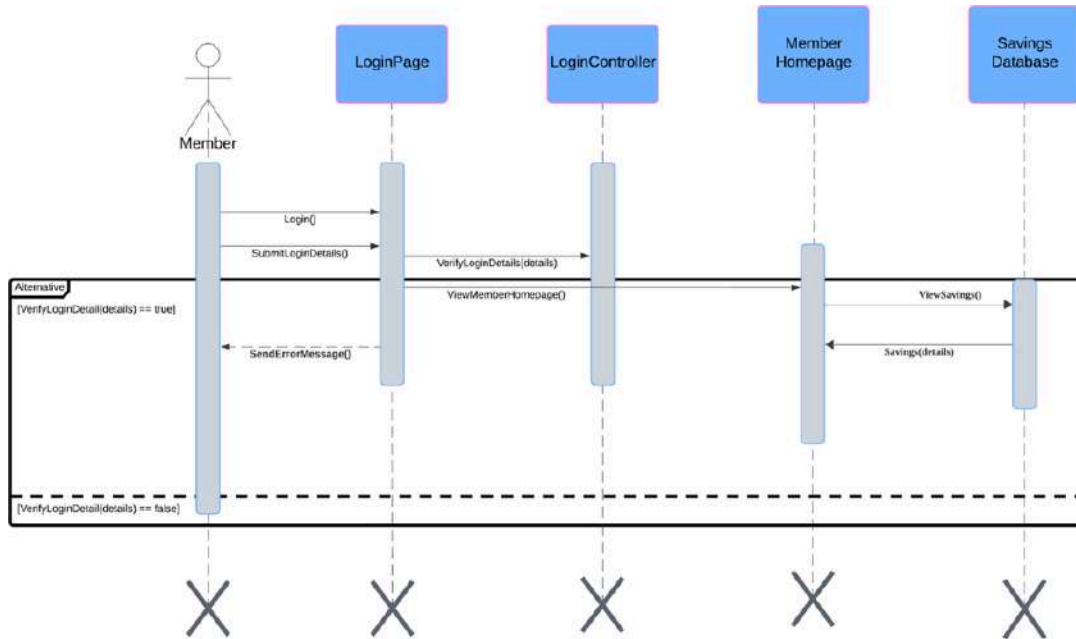


Figure 4: Sequence Diagram for Check Current Saving

5.3.1.3 UC23: Use Case Apply Membership Discontinuation

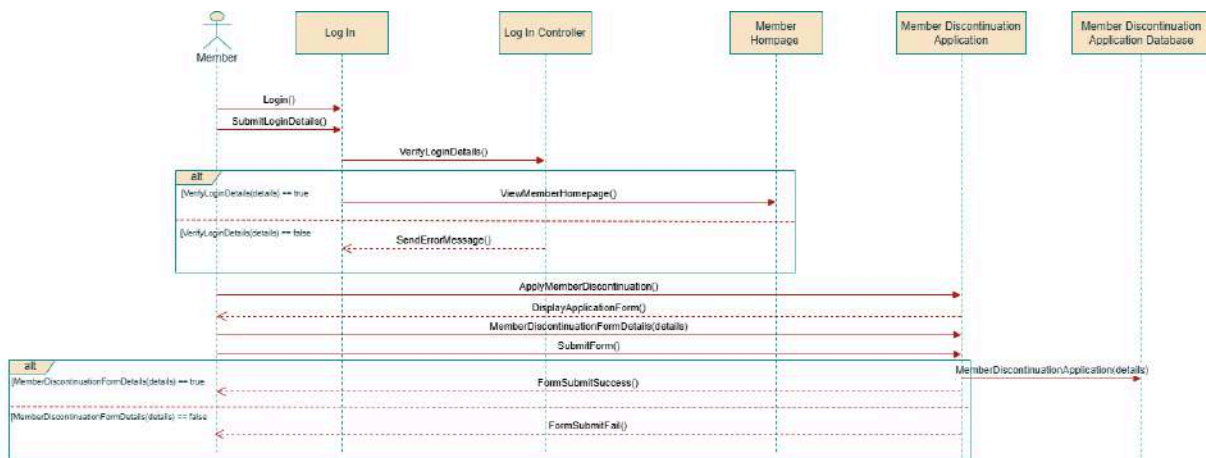


Figure 5: Sequence Diagram for Apply Membership Discontinuation

5.3.2 Module <Admin/Staff>

5.3.2.1 UC10: Use Case View Staff Menu

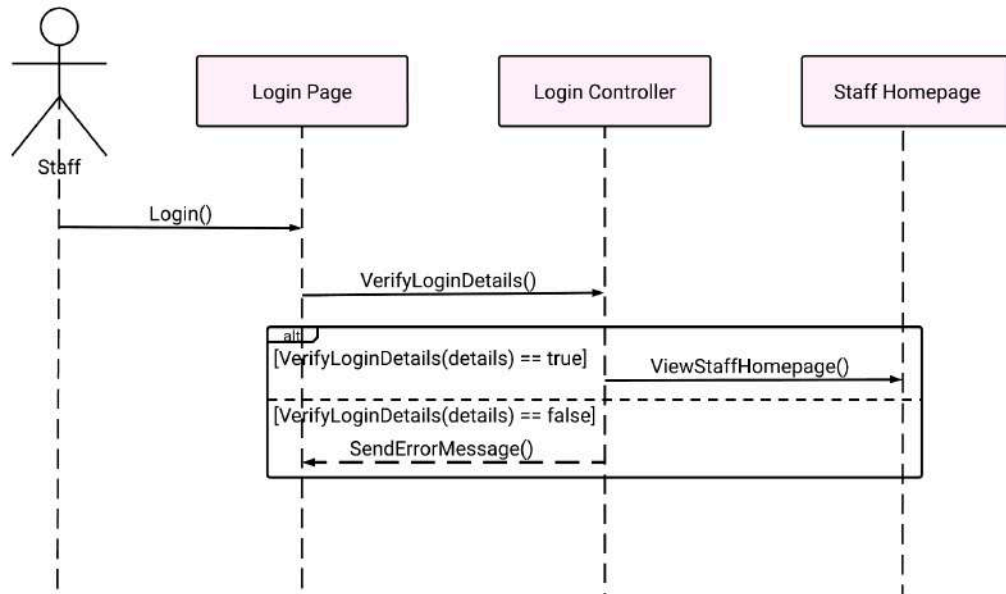


Figure 6: Sequence Diagram for View Staff Menu

5.3.2.2 UC06: Use Case View Member Application Form

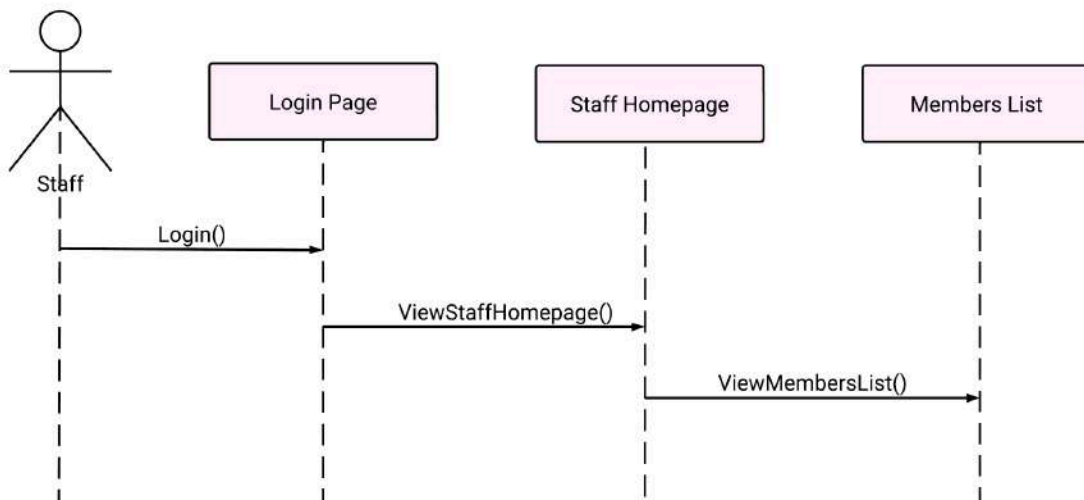


Figure 7: Sequence Diagram for View Member Application Form

5.3.2.3 UC07: Use Case Validate Member Application Form

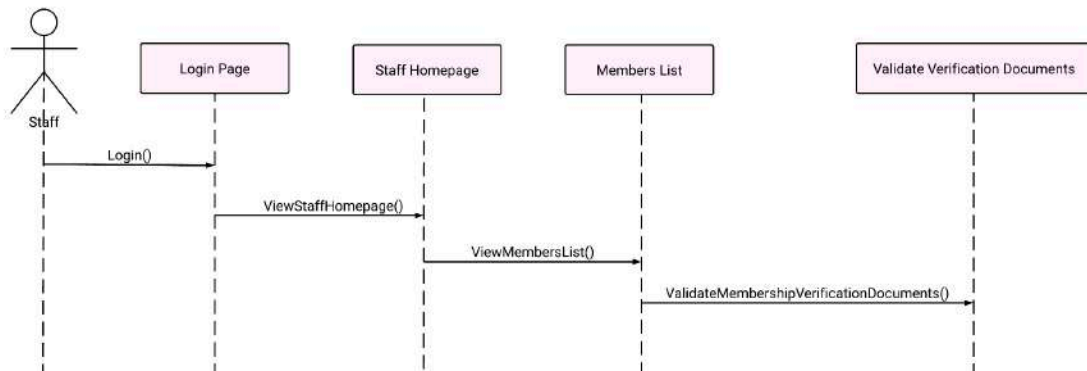


Figure 8: Sequence Diagram for Validate Member Application Form

5.3.2.4 UC09: Use Case View Loan Application Form

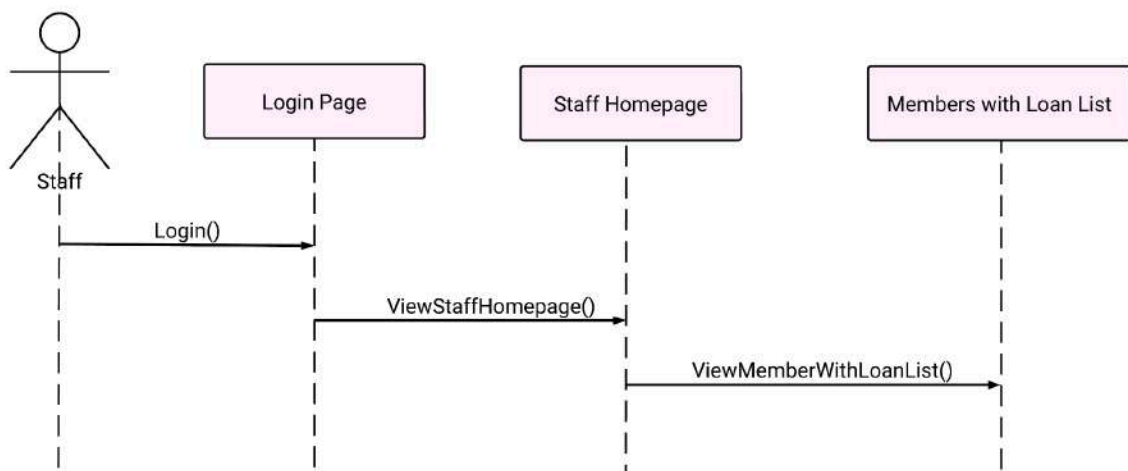


Figure 9: Sequence Diagram for View Loan Application Form

5.3.2.5 UC08: Use Case Validate Loan Application Form

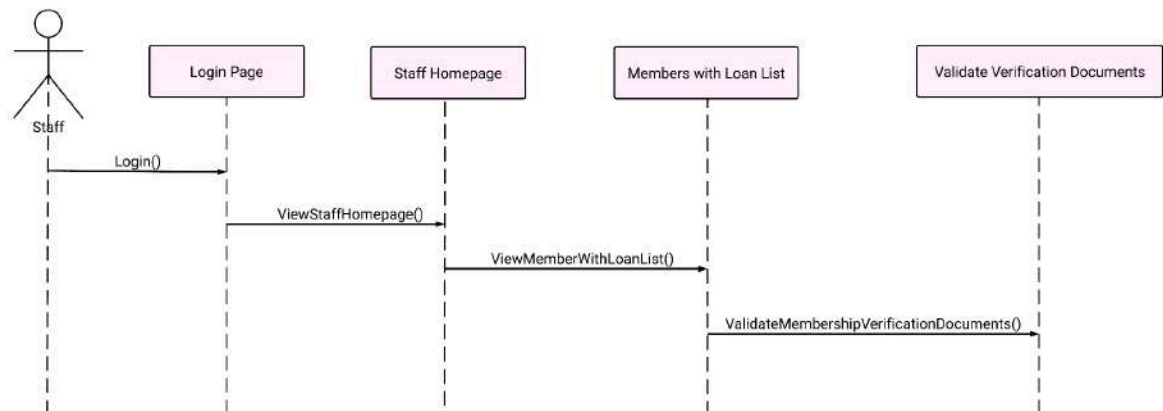


Figure 10: Sequence Diagram for Validate Loan Application Form

5.3.2.6 UC24: Use Case View Monthly Report

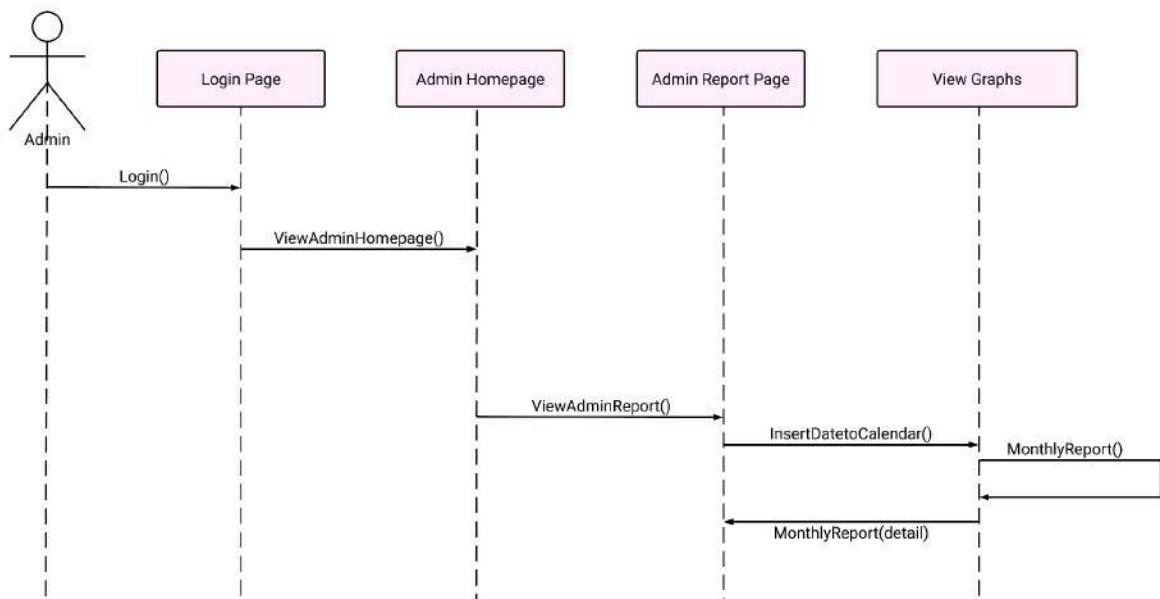


Figure 11: Sequence Diagram for View Monthly

5.3.2.7 UC25: Use Case View Annual Report

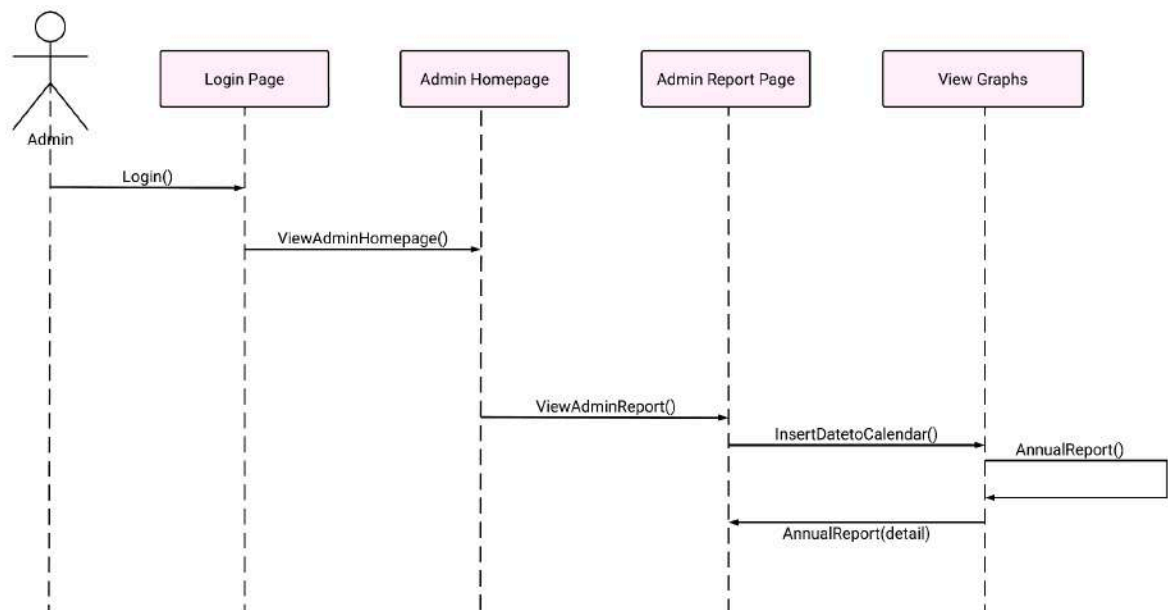


Figure 12: Sequence Diagram for View Annual Report

5.3.2.8 UC26: Use Case Add New Admin

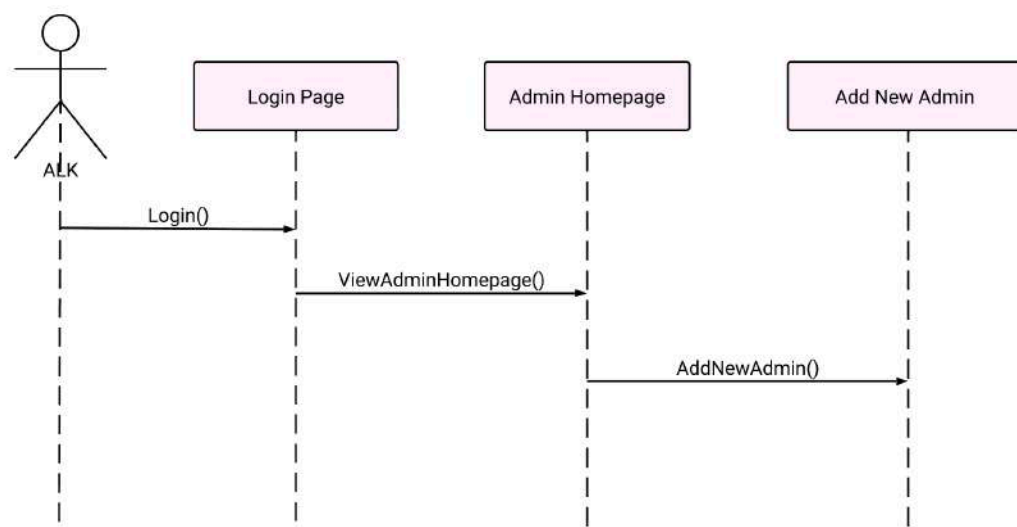


Figure 13: Sequence Diagram for Add New Admin

5.3.2.9 US27: Use Case Add New ALK

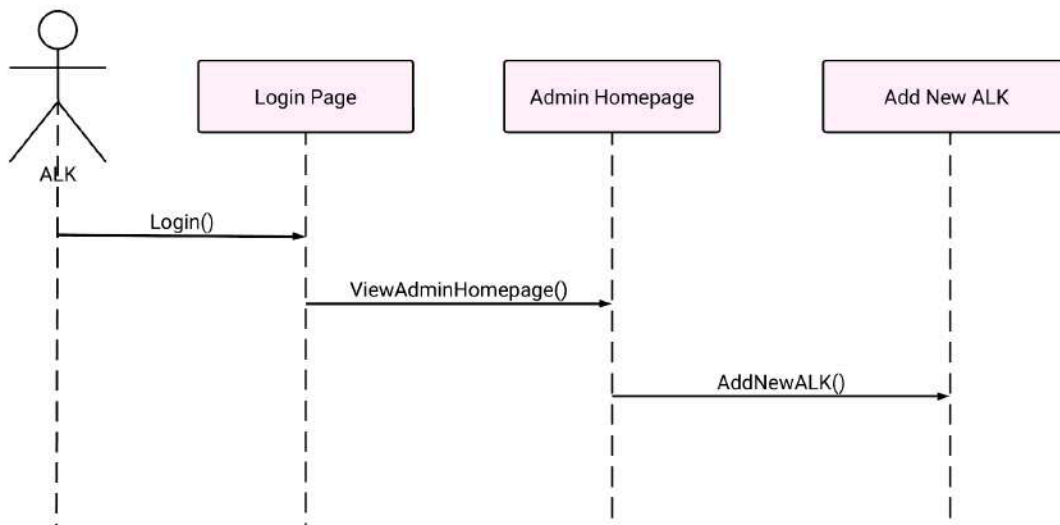


Figure 14: Sequence Diagram for Add New ALK

5.3.2.10 UC28: Use Case View Membership Discontinuation Form

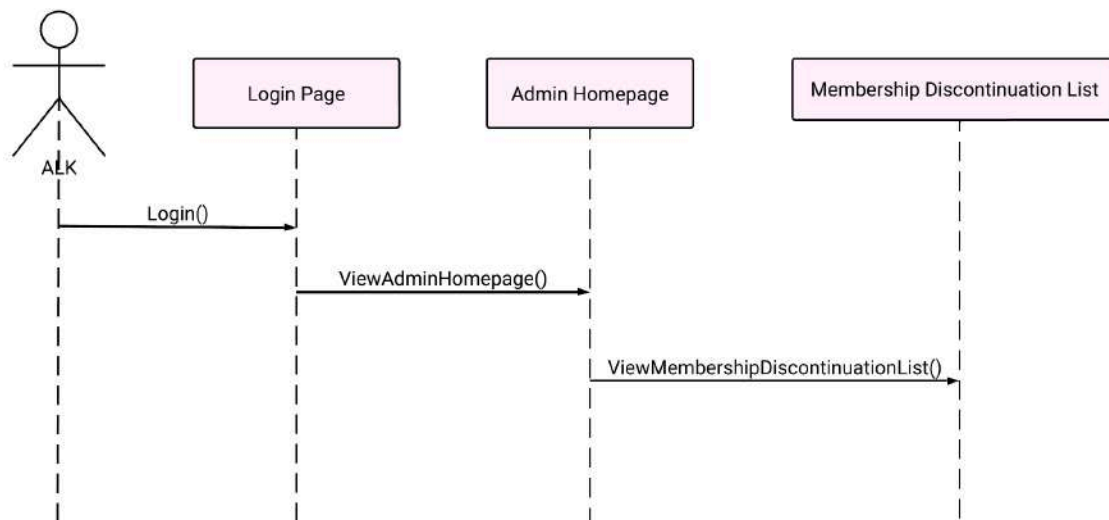


Figure 15: Sequence Diagram for View Membership Discontinuation Form

5.3.2.11 UC29: Use Case Approve Membership Discontinuation

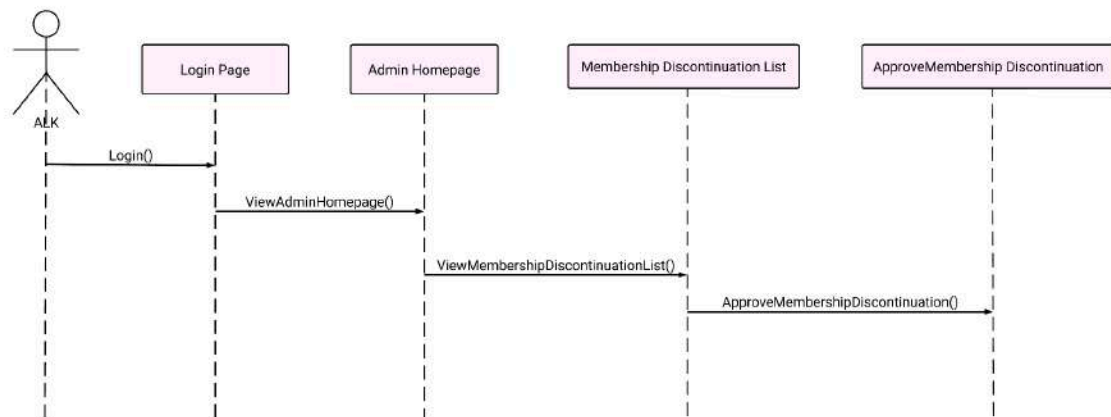


Figure 16: Sequence Diagram for Approve Membership Discontinuation Form

5.3.3 Module <ALK>

5.3.3.1 UC22: Use Case View ALK Menu

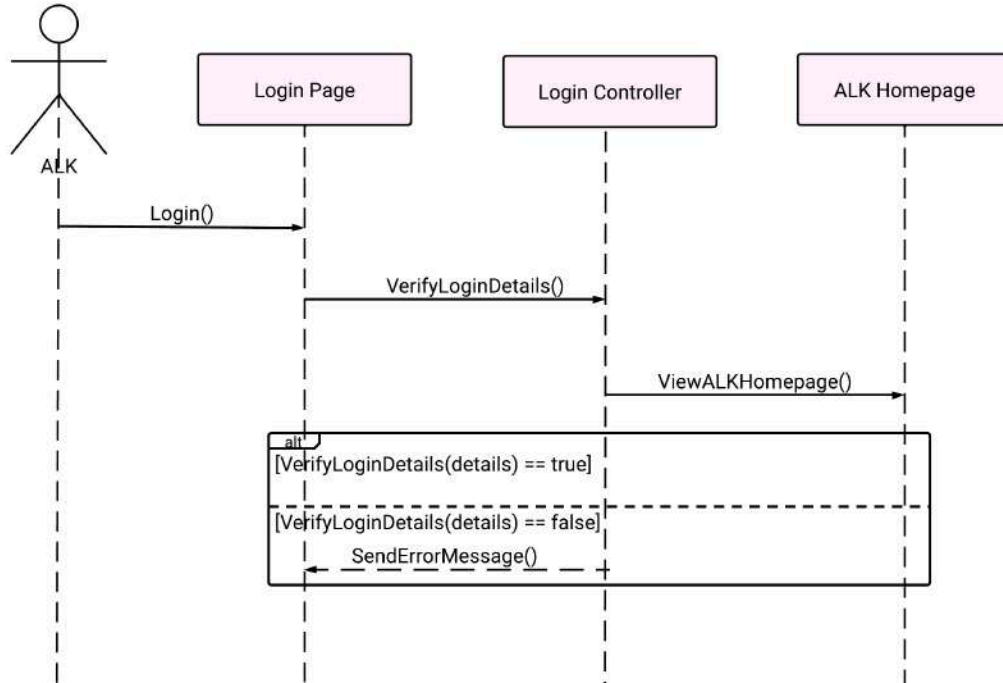


Figure 17: Sequence Diagram for View ALK Menu

5.3.3.2 UC21: Use Case View Validated Member Application Form

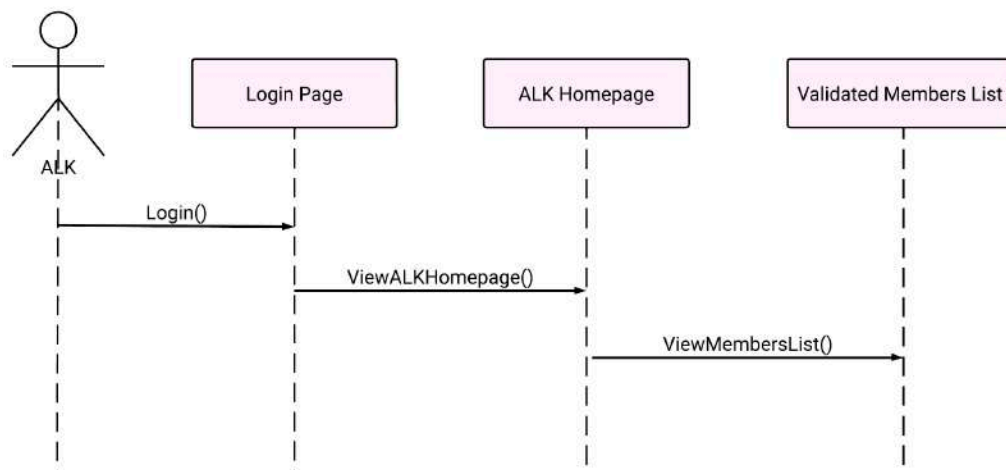


Figure 18: Sequence Diagram for View Validated Member Application Form

5.3.3.3 UC18: Use Case Approve Member Application

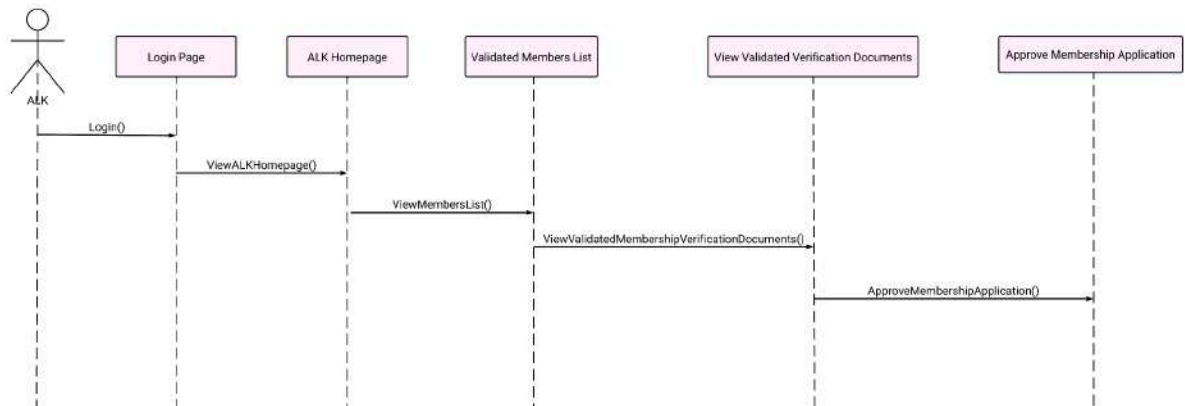


Figure 19: Sequence Diagram for Approve Member Application

5.3.3.4 UC20: Use Case View Validated Loan Application Form

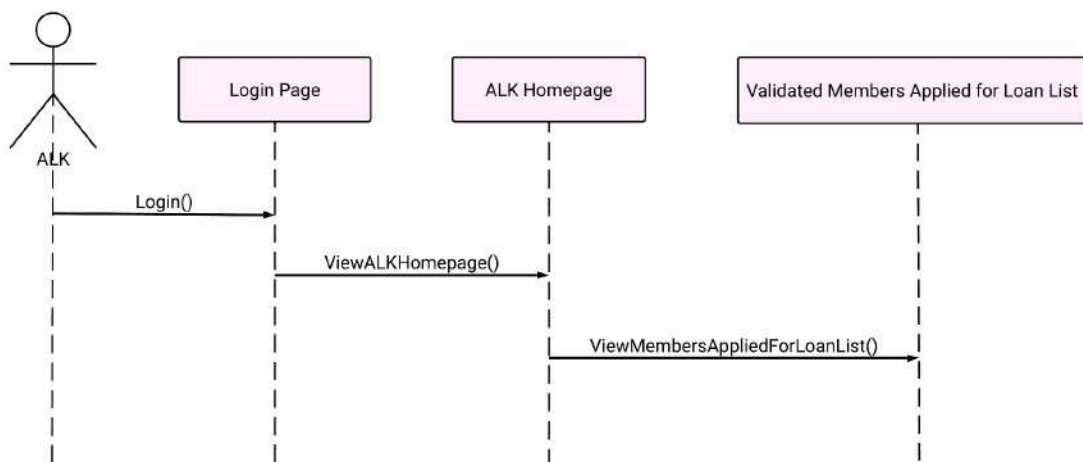


Figure 20: Sequence Diagram for View Validated Loan Application Form

5.3.3.5 UC19: Use Case Approve Loan Application

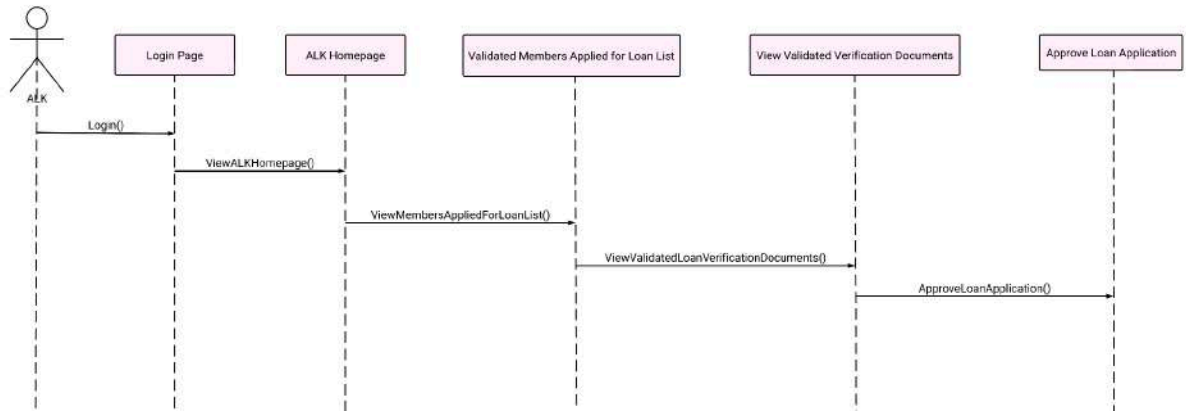


Figure 21: Sequence Diagram for Approve Loan Application

5.3.3.6 UC30: Use Case View Monthly Report

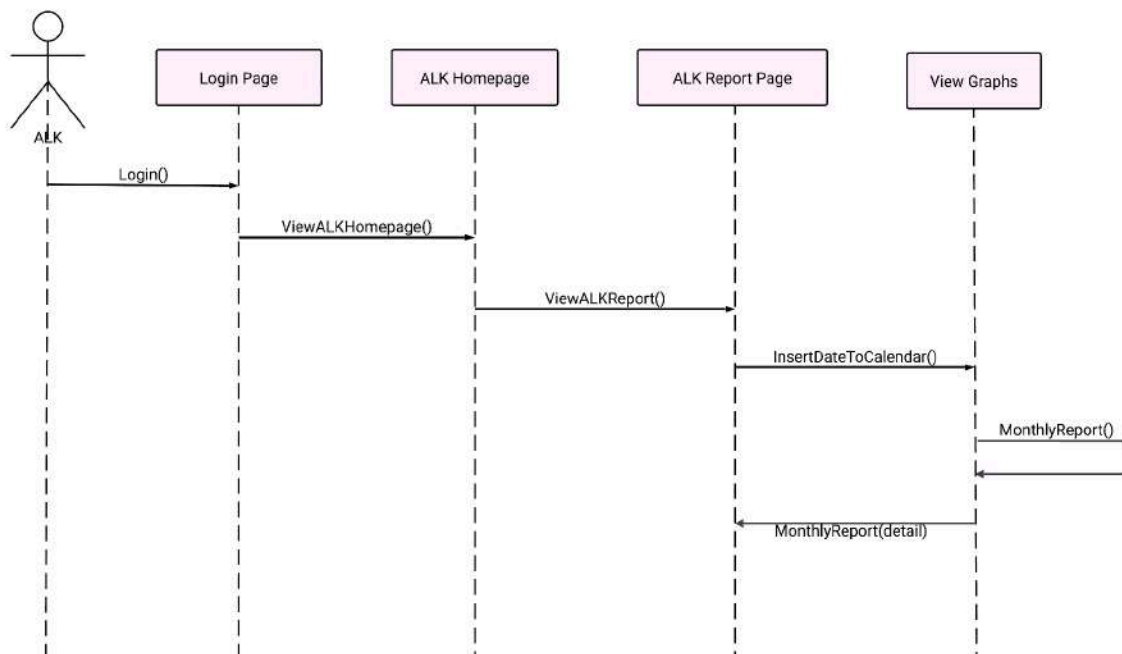


Figure 22: Sequence Diagram for View Monthly Report

5.3.3.7 UC31: Use Case View Annual Report

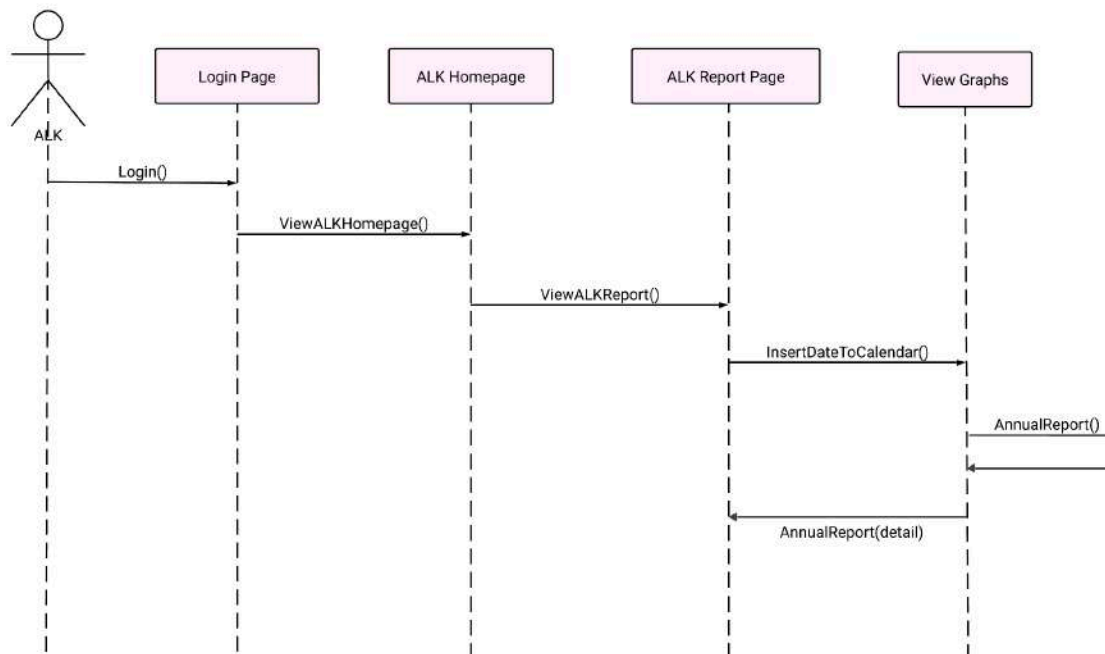


Figure 23: Sequence Diagram for View Annual Report

5.3.4 Module <Non-Member>

5.3.4.1 UC01: Use Case Apply Membership

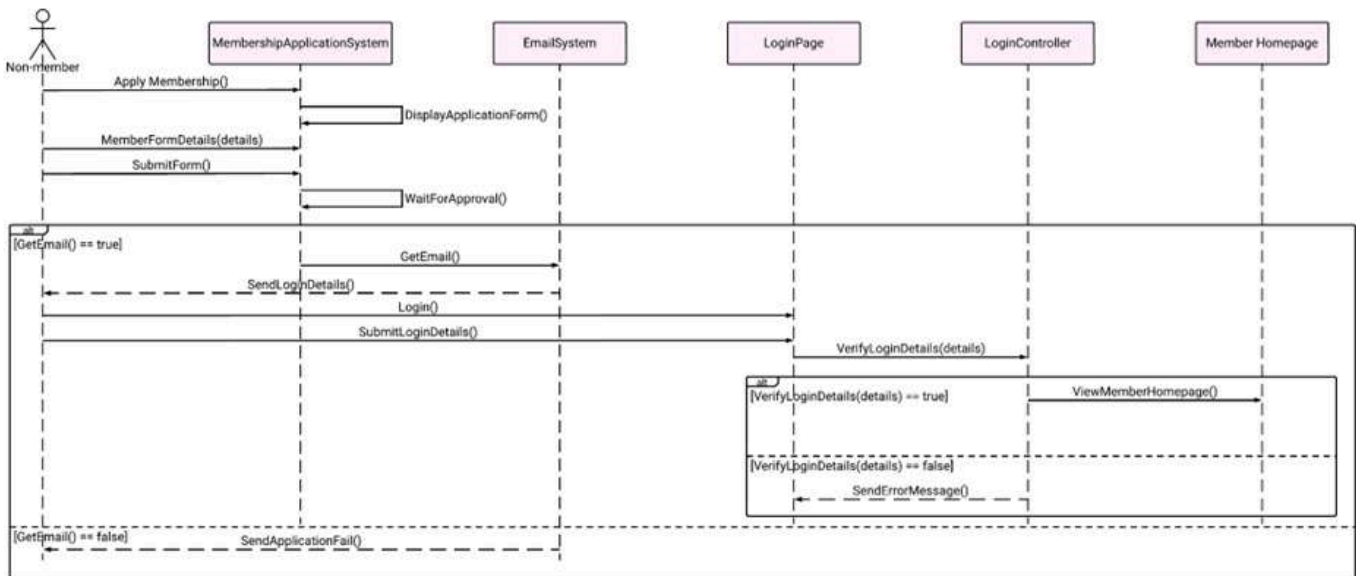


Figure 24: Sequence Diagram for Apply Membership

5.3.4.2 UC03: Use Case View Homepage

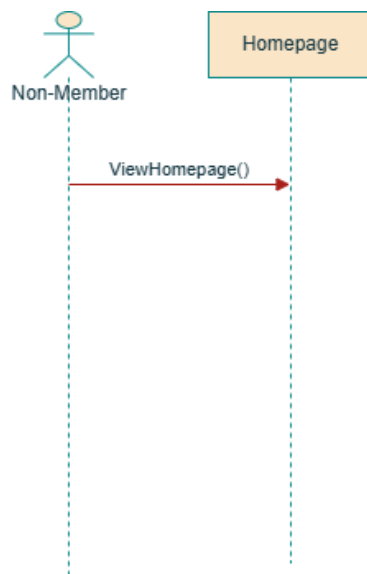


Figure 25: Sequence Diagram for View Homepage

5.3.4.3 UC02: Use Case Get Member Approval

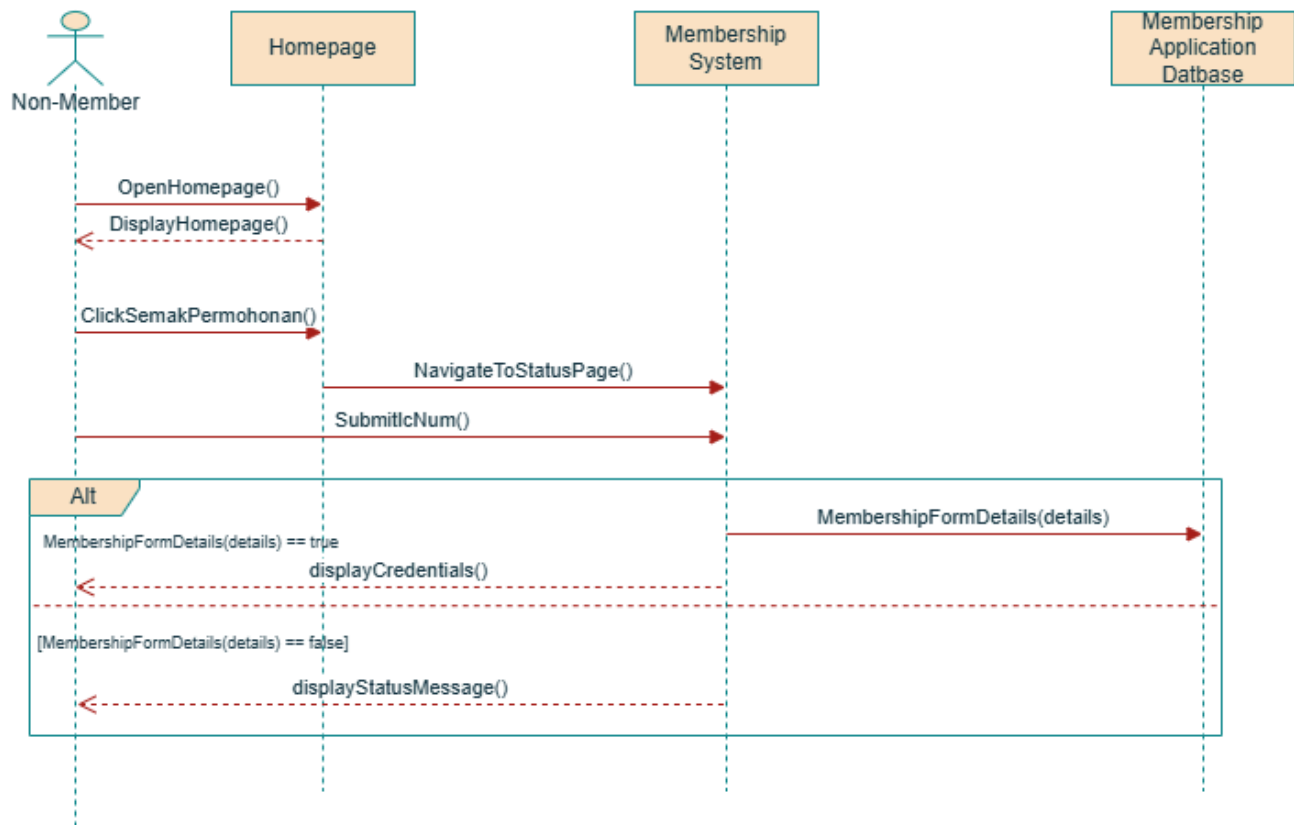


Figure 26: Sequence Diagram for Get Member Approval

5.3.5 Module <Authentication>

5.3.5.1 UC05: Use Case Login

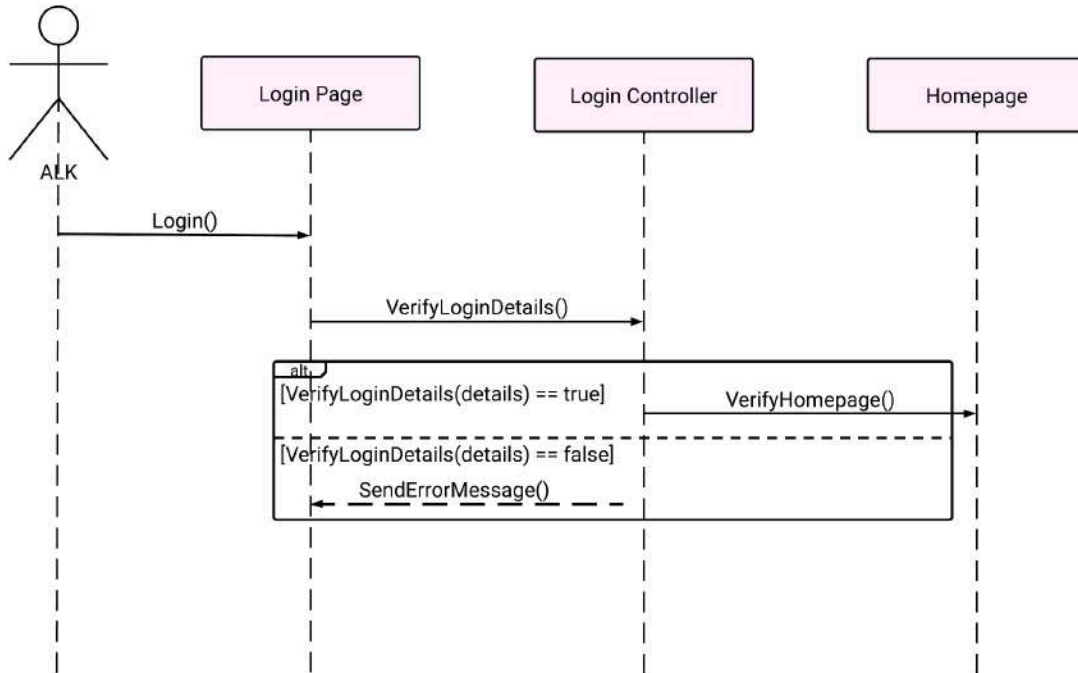


Figure 27: Sequence Diagram for Login

5.3.5.2 UC04: Use Case Log Out

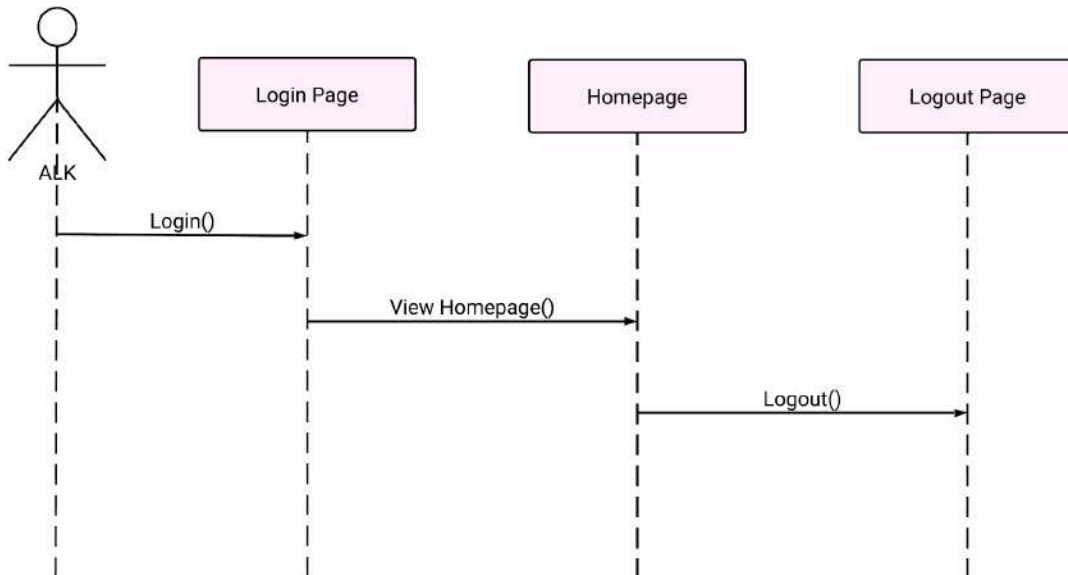


Figure 28: Sequence Diagram for Log Out

5.3.6 Module <Loan>

5.3.6.1 UC14: Use Case Apply Loan

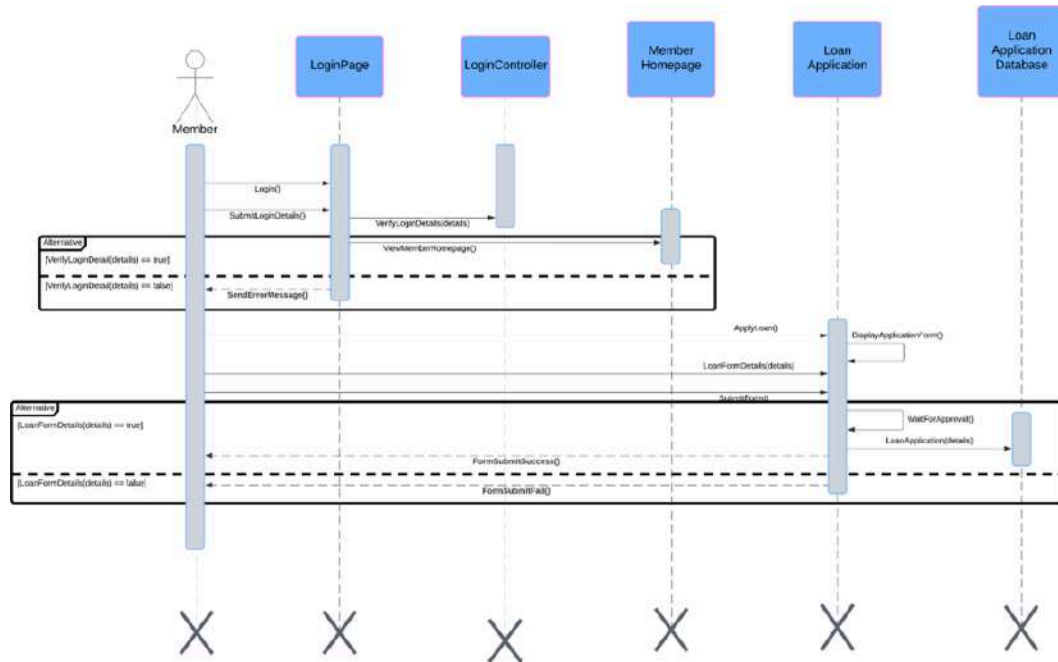


Figure 29: Sequence Diagram for Apply Loan

5.3.6.2 UC11: Use Case Check Loan Balance

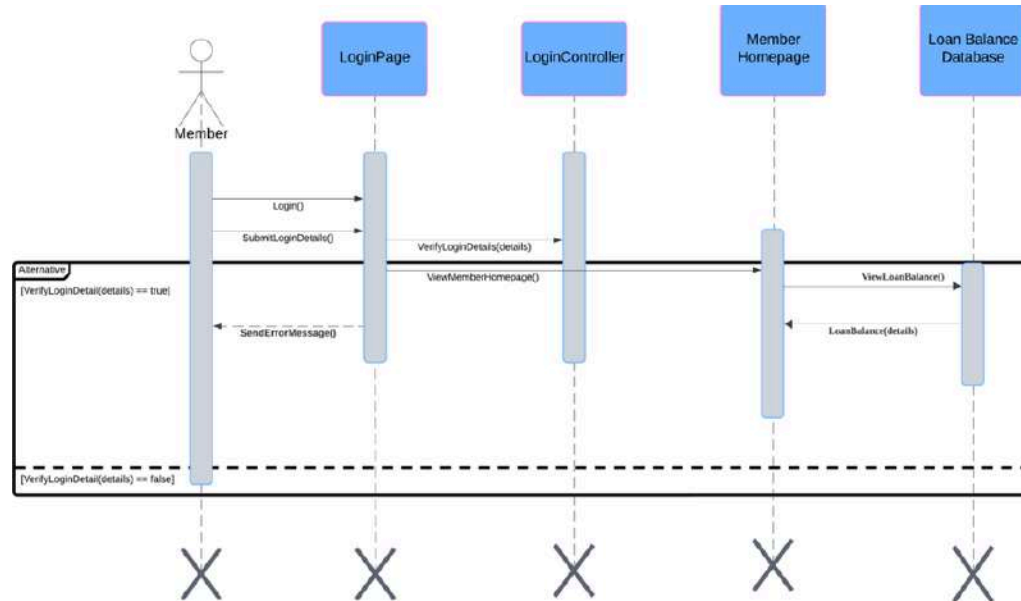


Figure 30: Sequence Diagram for Check Loan Balance

5.3.6.3 UC12: Use Case Check Loan Approval

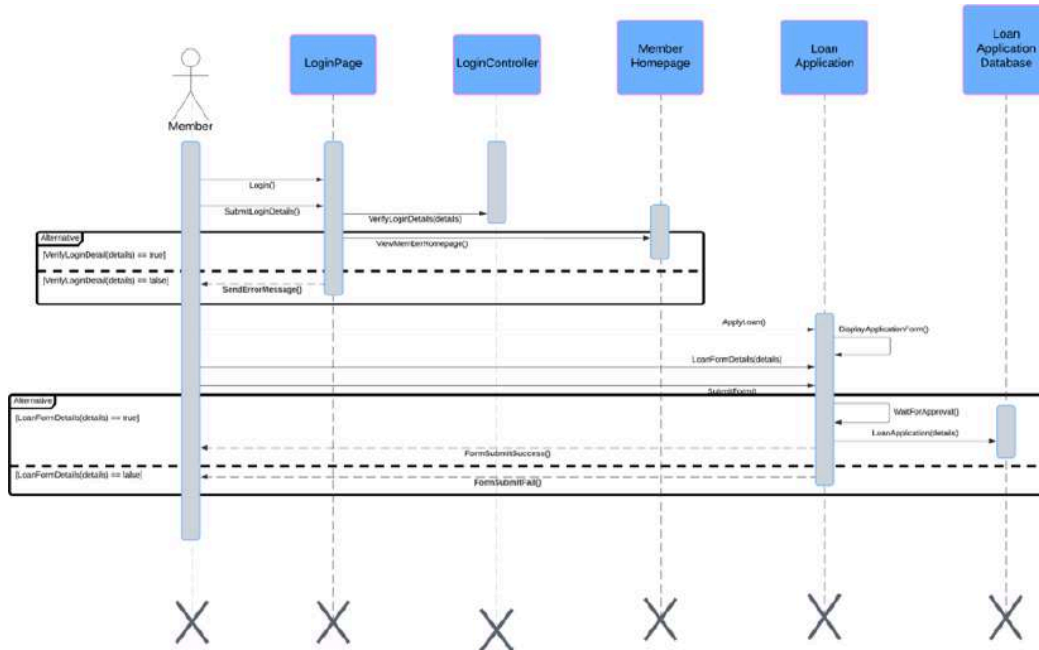


Figure 31: Sequence Diagram for Check Loan Approval

5.3.6.4 UC13: Use Case Find Transaction History

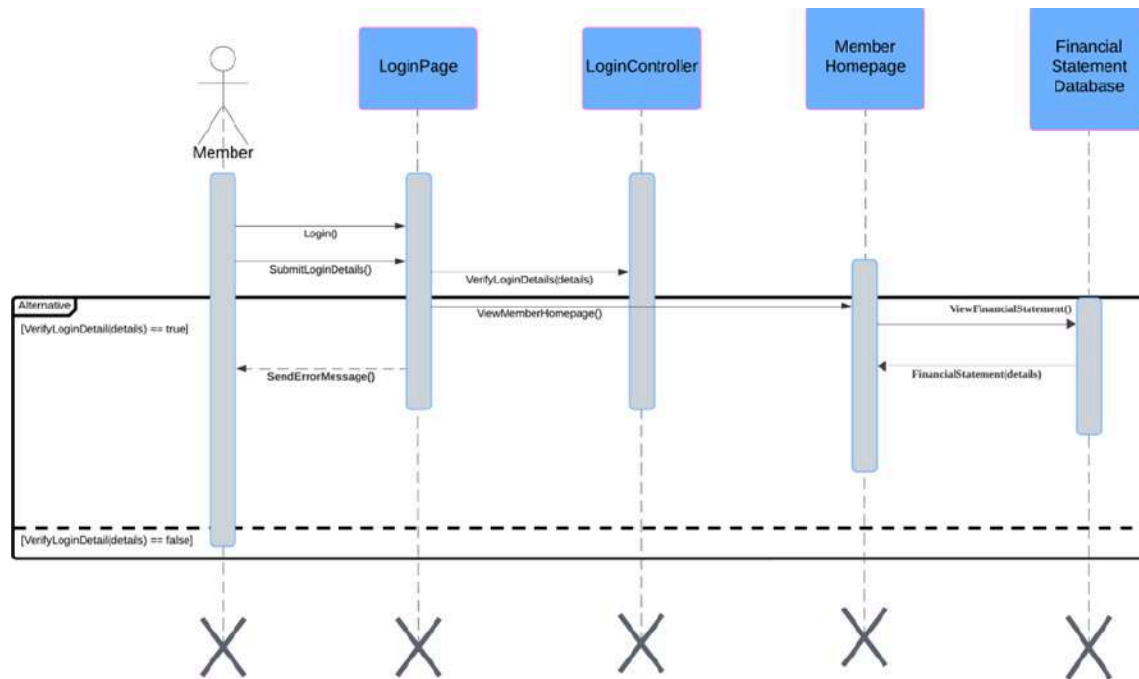


Figure 32: Sequence Diagram for Find Transaction History

6.0 Development Software, Language, Technology, API and Tools

Software

The softwares used while developing Koperasi KADA system are Visual Studio Code, Github, XAMPP Control Panel and phpMyAdmin. We used Visual Studio Code to do the coding part of the system. As we used several different languages in our coding, Visual Studio Code was the optimal software to be used as it supports many types of programming languages. As this was a group project, we linked our Visual Studio Code to Github, so that everyone could have access to the last updated code with ease. After every successful update, each member pushes the code to github and merges it to the existing code. Then, each member is required to pull from Github origin before beginning with their coding to have the updated code. XAMPP Control Panel is the software we used as a web server for the proposed system. XAMPP has multiple modules under its software, such as Apache, MySQL, FileZilla, Mercury and Tomcat. For our proposed system, only Apache and MySQL were used. Apache functions as the web server, enabling the execution of PHP scripts, which are developed using Visual Studio Code. phpMyAdmin was used in the process of the system development to handle the administration of MySQL database servers.

Language

We used multiple programming languages to have a successful functioning system. The programming languages used in the system development are Hypertext Preprocessor (PHP), Hypertext Markup Language (HTML), Cascading Style Sheets (CSS) and Structured Query Language (SQL). The functionalities of the system were developed using PHP language as we used the Model-View-Controller (MVC) architecture. The system was styled mainly using HTML language and CSS language for a few interfaces under the View of MVC architecture. Lastly, SQL language was used for the database declaration and modification of the system.

Technology

The technology we have used in this system development is that we have used Hostinger as our web hosting provider. The system that we developed using Visual Studio Code, Github and phpMyAdmin was pushed to Hostinger to have a website that connected to the database server.

Tools

The tools that we used for the documentation of this system are lucidchart and draw.io. Only the use case diagram of this system was developed using draw.io whereas the rest of the diagrams were developed using lucidchart.

7.0 Function/Module Description

7.1 Authentication module

Person-in-charge: Harini A/P Sangaran

Module Description:

The Authentication Module manages interactions between members, staff and ALK with the login page. The module checks the authenticity of the user ID and password of each member, staff or ALK. If it's authenticated, then the members, staff or ALK can login to the page and interact with the system.

Use Cases:

1. Login (UC05): This use case allows users to access the system.
2. Logout (UC04): This use case allows users to exit the system.

7.2 Non-member Module

Person-in-charge: Nurul Ika Syafiny Binti Azhar

Module Description:

The Non-member Module manages interactions with individuals who are not yet registered members. It facilitates the application process, allows non-members to view the homepage, and provides the necessary steps for membership approval.

Use Cases:

1. Apply for Membership (UC01): Allows a non-member to submit an application to become a member.
2. View Homepage (UC02): Enables non-members to access a public-facing homepage with limited content.
3. Get Membership Approval (UC03): Check status of membership application that has been applied.

7.3 Member Module

Person-in-charge: Nuraisyah Binti Mohd Zikre

Module Description

In this module, it manages individuals who are now a qualified member of Koperasi KADA after applying for membership and they are required to log in into the system in order to gain access to the system. It allows members to view savings, get invoices and view member menus.

Use Cases:

1. Get invoice (UC 15): Allows member to view and get their invoice of every transaction made
2. Check current saving share (UC 16): Allows user to view their current savings and share
3. View member menu (UC 17): Allows member to view the menu where the menu is the function that accessible only for a member of Koperasi KADA
4. Apply membership termination (UC 23): Allows member to terminate the membership with Koperasi KADA by submit the application

7.4 Loan Module

Person-in-charge: Nurul Ika Syafiny Binti Azhar and Nuraisyah Binti Mohd Zikre

Module Description

In this module, it allows the members of Koperasi KADA to manage loan related activities which consists of applying for a loan, checking loan balance, view loan approval (if any) and transaction history. Where they are required to log in into the system in order to get the access.

Use Cases:

1. Check Loan Balance (UC11): It allows member of Koperasi KADA to view outstanding balance of loan they applied
2. Check Loan Approval (UC12): It allows members of Koperasi KADA to be notified with the status of loan application that they applied
3. Find Transaction History (UC13): It allows member of Koperasi KADA to view the transaction record of every transaction they received
4. Apply for Loan (UC14): It allows the member of Koperasi KADA to apply for a loan

7.5 Staff Module

Person-in-charge: Harini A/P Sangaran and Damiya Aina binti Basir Abd Shammad

Module Description

The Staff Module manages interactions with Koperasi KADA staff by providing access to system menus, approving applications, and managing data validation. It supports essential staff operations, ensuring smooth administrative processes.

Use Cases:

1. View staff menu (UC10): Allows staff to see the menu of the system where all their functions are available.
2. Validate member application form (UC07): Validates each submitted membership application form and ensures the details are true before being sent to the ALK.
3. View member application form (UC 06): Allows staff members to view submitted membership application applied by non-member
4. Validate loan application form (UC08): Validates each submitted loan application form and ensures the details are true before being sent to the ALK.
5. View loan application form (UC 09): Allows staff members to view submitted loan application applied by member
6. View member discontinuation form (UC 28): Allows staff members to view submitted member termination application
7. Approve membership discontinuation (UC 29): Staff will approved or disapproved the membership termination request submitted by member of Koperasi KADA
8. Add new admin (UC 26): Add another admin to manage the process in Koperasi KADA
9. Add new ALK (UC 27): Add another ALK to manage the process in Koperasi KADA
10. View monthly report (UC 24): View the monthly generated report
11. View annual report (UC 25): View the annual generated report

7.6 ALK Module

Person-in-charge: Damiya Aina binti Basir Abd Shammad

Module Description

The ALK Module manages tasks and approvals performed by ALK members. It allows viewing and validating member and loan applications, approving or rejecting submissions, and accessing financial transaction data for informed decision-making.

Use Cases:

1. Approve member application (UC18): Allows ALK members to approve or reject membership applications.
2. Approve loan application (UC19): Enables ALK members to approve or reject loan applications.
3. View validated loan application form (UC20): Allows ALK members to view submitted and validated loan application forms.
4. View validated member application form (UC21): Enables ALK members to view submitted and validated membership forms.
5. View ALK menu (UC22): Provides ALK members with access to their menu and related functionalities.
6. View monthly report (UC 30): View the monthly generated report
7. View annual report (UC 31): View the annual generated report

8.0 System Credentials (Hosting)

8.1 Member

- ID number : 0008
- Password : 1234

8.2 Admin

- ID number : 0001
- Password : 1234

8.3 Ahli Lembaga Koperasi (ALK)

- ID number : 0001
- Password : 1234

9.0 Localhost Setup

1. Download and install XAMPP Control Panel from <https://www.apachefriends.org/>.
2. Start Apache and MySQL.

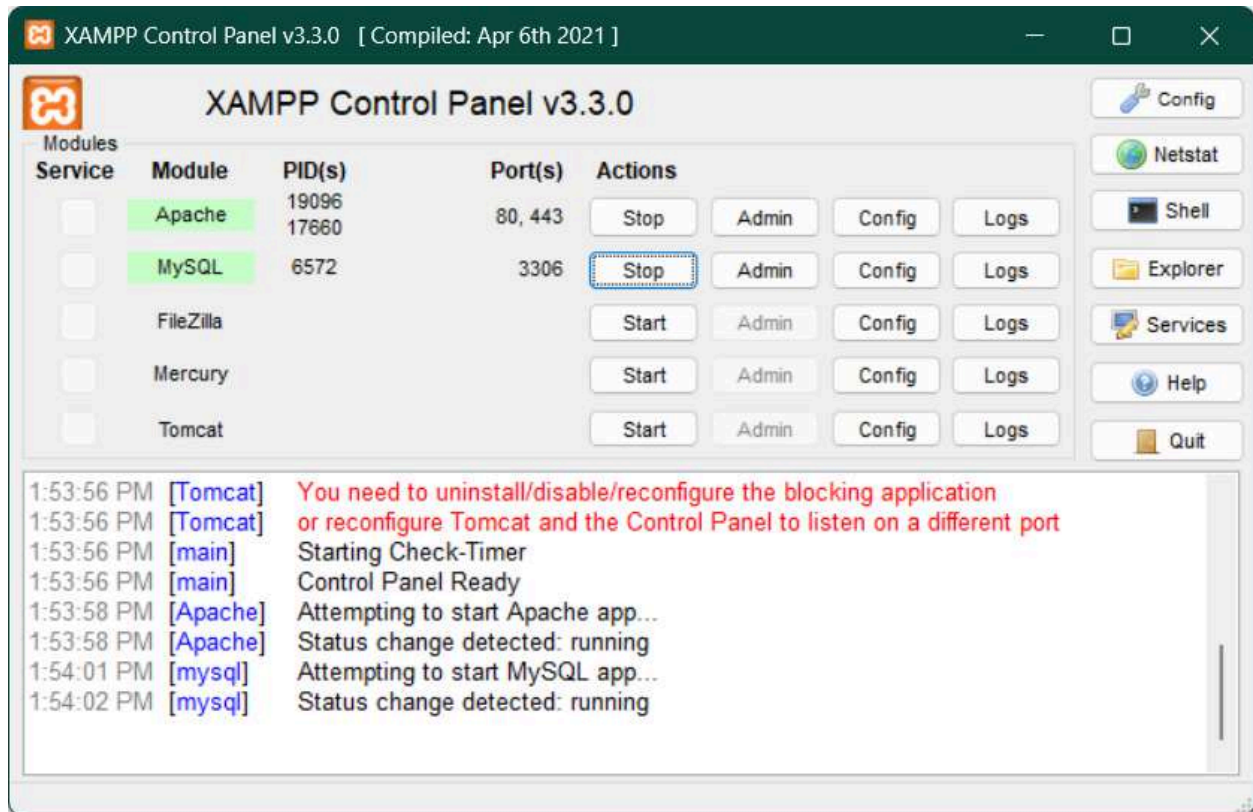


Figure 33: XAMPP Control Panel

3. Click on 'Admin' button under MySQL to go to phpMyAdmin page.
4. Create a database named 'koperasikada'.

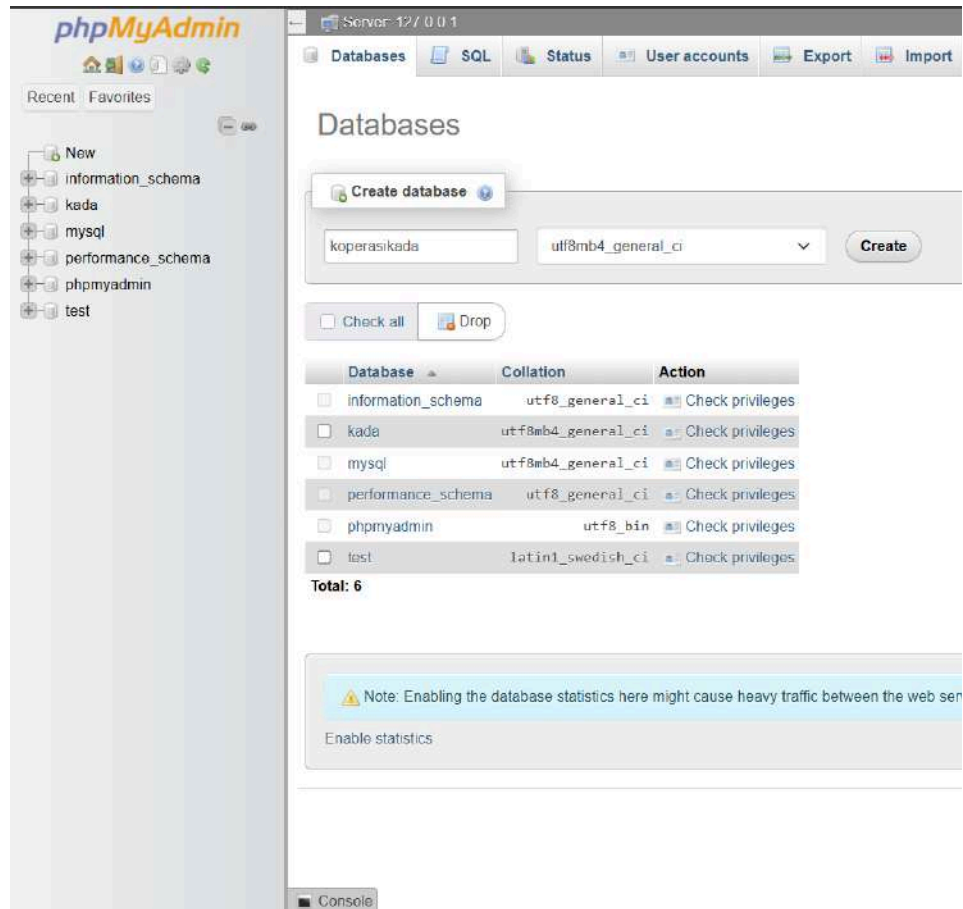


Figure 34: List of Database

5. Enter the SQL for each table under the SQL column of koperasikada database to establish databases.

10.0 System Interface for All Users

10.1 Non-Member user

10.1.1 Homepage Koperasi KADA



Figure 35: Homepage Interface

10.1.2 Apply Membership

Maklumat Pemohon

Name		Nombor K/P:	
Email			
Taraf Perkahwinan	Jantina	Agama	
Bangsa	Alamat Rumah		
Poskod	Negeri	Bandar	
Maklumat Pejabat			
No. Anggota		No. PF	
Jawatan & Gred		Alamat Pejabat	
Poskod		No. Tel/Fax	

Contact Details

No. Tel Bimbit

No. Tel Rumah

Maklumat Keluarga

Nama Ahli Keluarga 1

Hubungan

No. Kad Pengenalan

Nama Ahli Keluarga 2

Hubungan

No. Kad Pengenalan

Gaji Bulanan

Perkara	RM
Yuran Masuk	
Modal Syer	

Perkara	RM
Yuran Masuk	
Modal Syer	
Modal Yuran	
Wang Deposit Anggota	
Sumbangan Tabung Kebajikan (Al-Abrar)	
Simpanan Tetap	
Lain-lain	

Simpan Maklumat

Figure 36: Apply Membership Interface

10.1.3 Check Membership Application Status

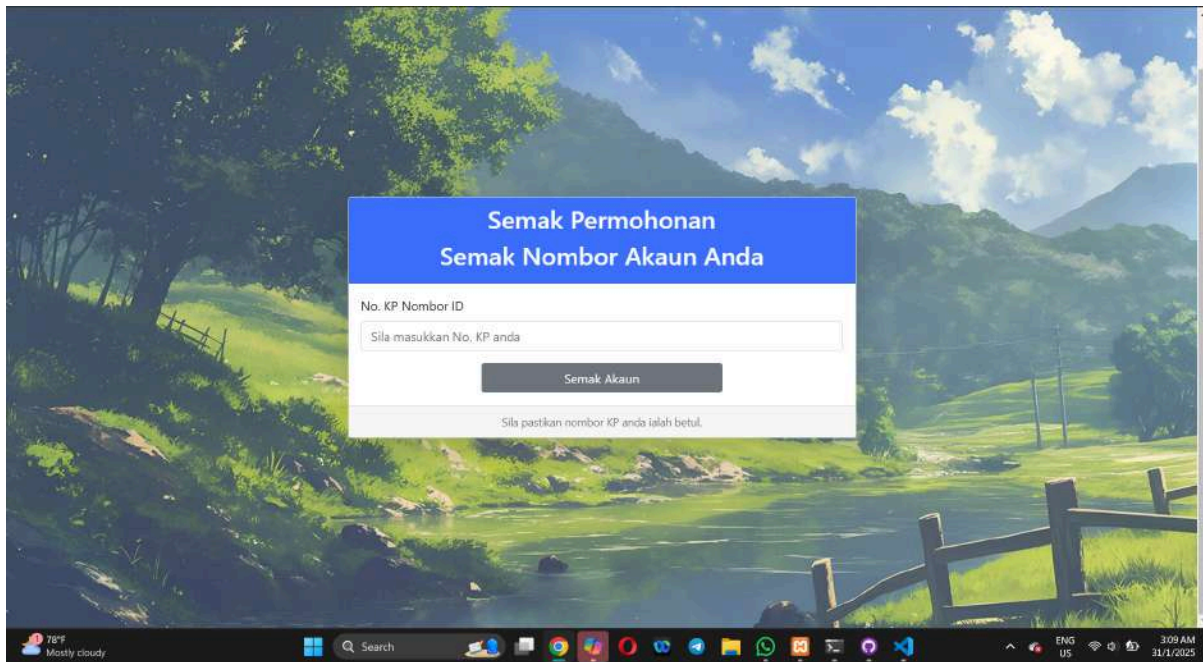


Figure 37: Check Membership Application Interface

10.1.4 Approved Membership Status

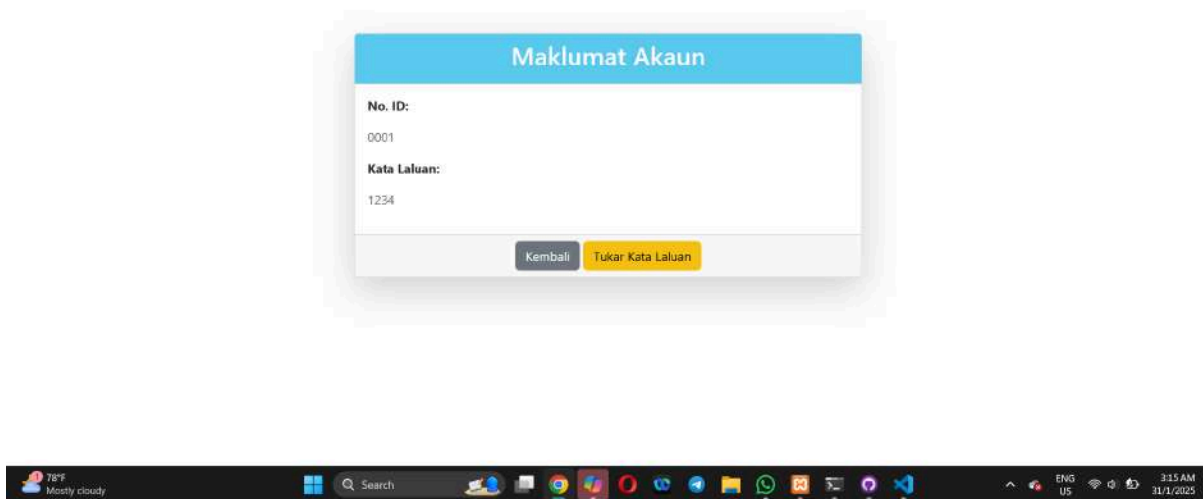


Figure 38: Approved Membership Status Interface

10.1.5 Pending Membership Status



Figure 39: Pending Membership Status Interface

10.1.6 Reviewed Membership Status



Figure 40: Reviewed Membership Status Interface

10.1.7 Rejected Membership Status

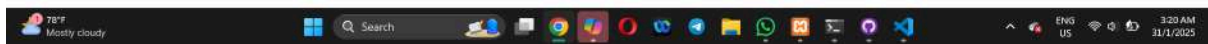


Figure 41: Rejected Membership Status Interface

10.2 Member User

10.2.1 Login Page

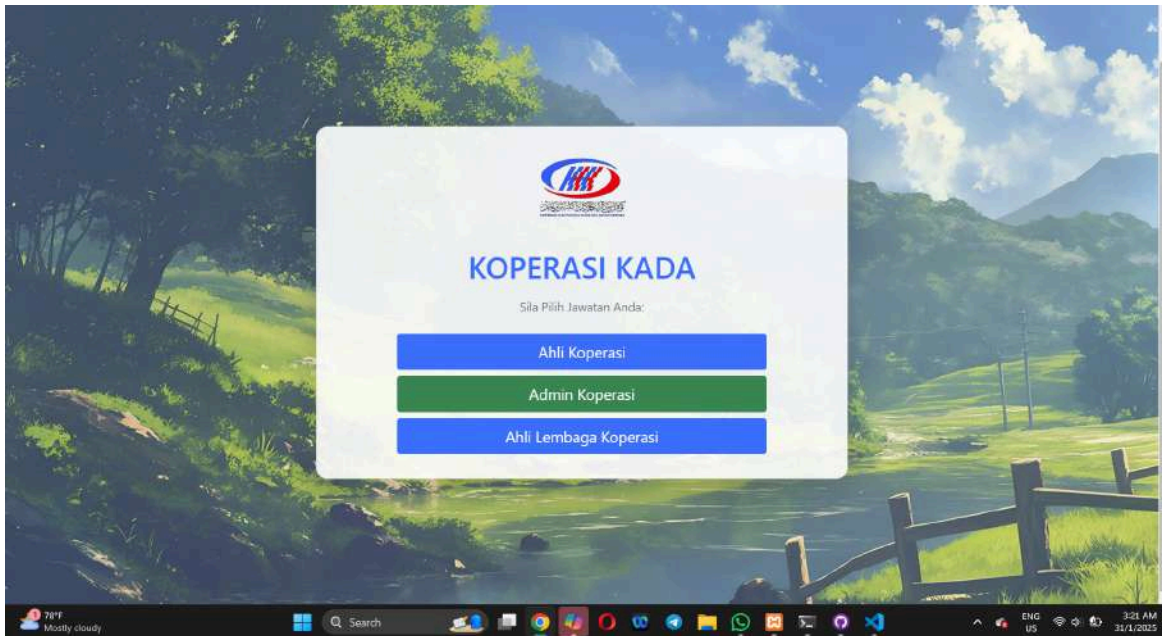


Figure 42: Login Page Interface

10.2.2 Member Login

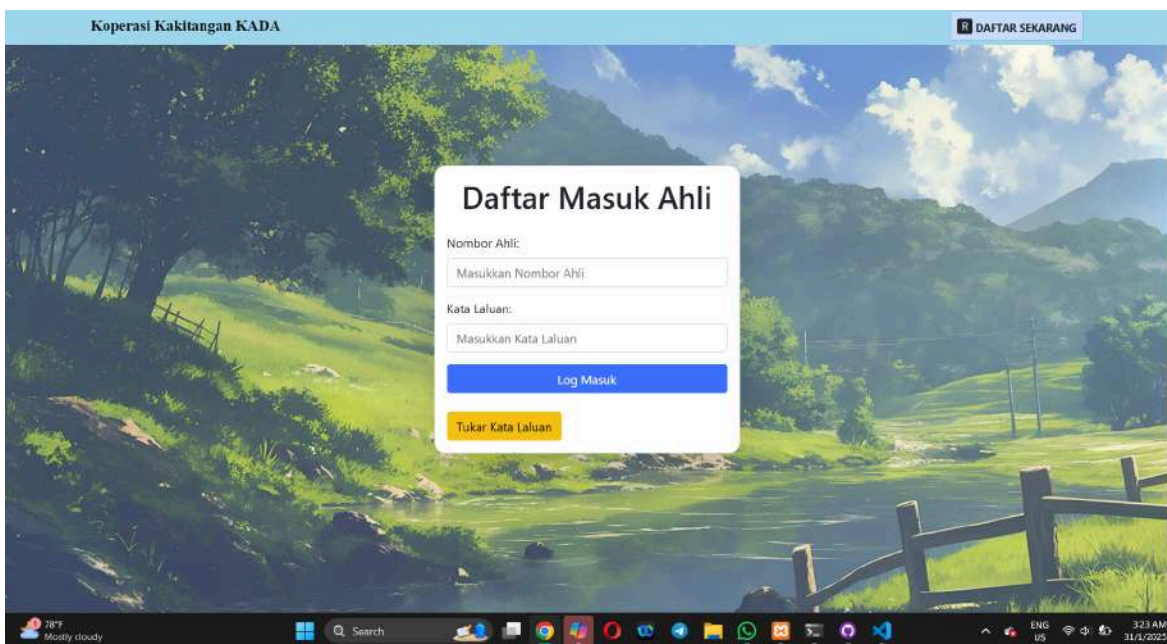


Figure 43: Member Login Interface

10.2.3 Member Homepage



Figure 44: Member Homepage Interface

10.2.4 Member Details

The screenshot displays the 'MAKLUMAT AHLI' (Member Details) form. The header includes the organization's name, a 'Pilihan' dropdown, and a 'DAFTAR KELUAR' button. The form is divided into two main sections: 'Maklumat Peribadi' (Personal Information) and 'Maklumat Pejabat' (Office Information).

Maklumat Peribadi

Name	Harini A/P Sangaran		Nombor K/P:	041201070330	
Email	harini.sangaran@gmail.com				
Taraf Perkahwinan	Jantina	Agama			
Bujang	Perempuan	Hindu			
Bangsa	India		Alamat Rumah		
			21, Lorong Impian Indah 2, Taman Impian Indah		
Poskod	Negeri	Bandar			
14000	Pulau Pinang	Bukit Mertajam			

Maklumat Pejabat

No. Anggota	No. PF	
0081	5678	
Jawatan & Gred	Alamat Pejabat	
A	MA6 KTDI	

The Windows taskbar at the bottom shows the date as 31/1/2025 and the time as 3:42 AM.

Koperasi Kakitangan KADA

Pilihan

DAFTAR KELUAR

Gaji Bulanan

Gaji Bulanan

5000.00

Yuran & Sumbangan

Perkara	RM
Yuran Masuk	55.00
Modal Syer	0.00
Modal Yuran	0.00
Wang Deposit Anggota	0.00
Sumbangan Tabung Kebajikan (Al-Abrar)	0.00
Simpanan Tetap	0.00
Lain-lain	0.00

Kemaskini Maklumat

Figure 45: Member Details Interface

10.2.5 Update Member Details

Koperasi Kakitangan KADA

Pilihan

DAFTAR KELUAR

KEMASKINI MAKLUMAT PEMOHON

Maklumat Peribadi

Name

Harini A/P Sangaran

Nombor K/P:

041201070330

Email

harini.sangaran@gmail.com

Taraf Perkahwinan

Bujang

Jantina

Perempuan

Agama

Hindu

Bangsa

India

Alamat Rumah

21, Lorong Impian Indah 2, Taman Impian Indah

Poskod

14000

Negeri

Pulau Pinang

Bandar

Bukit Mertajam

Maklumat Pejabat

No. Anggota

0081

No. PF

5678

Jawatan & Gred

A

Alamat Pejabat

MA6 KTDI

Koperasi Kakitangan KADA
Pilihan
DAFTAR KELUAR

No. Kad Pengenalan
750502025500

Gaji Bulanan
5000.00

Yuran & Sumbangan

Perkara	RM
Yuran Masuk	55.00
Modal Syer	0.00
Modal Yuran	0.00
Wang Deposit Anggota	0.00
Sumbangan Tabung Kebajikan (Al-Abrar)	0.00
Simpanan Tetap	0.00
Lain-lain	0.00

SIMPAN MAKLUMAT

Figure 46: Update Member Details Interface

10.2.6 Apply Loan

Koperasi Kakitangan KADA
Pilihan
DAFTAR KELUAR

JADUAL PEMBAYARAN BALIK PEMBIAYAAN

Tempoh (Tahun)	1	2	3	4	5	6
Tempoh (Bulan)	12	24	36	48	60	72
Jumlah Pembayaran	Ansuran Bulanan					
1,000.00	86.83	45.17	31.28	24.33	20.17	17.39
2,000.00	173.67	90.33	62.56	48.67	40.33	34.78
3,000.00	260.50	135.50	93.83	73.00	60.50	52.17
4,000.00	347.33	180.67	125.11	97.33	80.67	69.56
5,000.00	434.17	225.83	156.39	121.67	100.83	86.94
6,000.00	521.00	271.00	187.67	146.00	121.00	104.33
7,000.00	607.83	316.17	218.94	170.33	141.17	121.72

BORANG PERMOHONAN PINJAMAN

Jenis Pinjaman

BORANG PERMOHONAN PINJAMAN

Jenis Pinjaman
Sila Pilih Jenis Pinjaman

Jika 'Lain-lain', Nyatakan
Nyatakan Jenis Pinjaman Lain

Jumlah Pinjaman (RM)
Masukkan Jumlah Pinjaman

Tempoh Bayaran Balik (bulan)
Nyatakan Tempoh Bayaran Balik Bulanan

Ansuran Bulanan (RM)
Masukkan Ansuran Bulanan

Nama Bank
Nyatakan Nama Bank

Nombor Akaun
Masukkan Nombor Akaun

Muat Turun Templat Dokumen Borang Permohonan Pinjaman

Muat Naik Dokumen Lengkap
Choose File: No file chosen

HANTAR PERMOHONAN

Figure 47: Apply Loan Interface

10.2.7 Check Loan Application Status

SEMAK KELULUSAN PINJAMAN

Permohonan Anda Berjaya!

No. ID Member: 1
Nama Bank: RHB Bank
No. Akaun Bank: 1571003
Jenis Pinjaman: Skim Khas
Jumlah Pinjaman: 7000.00
Tempoh Pembayaran Pinjaman: 48
Ansuran Bulanan: 170.33

Permohonan Anda Sedang Diproses
Sila tunggu keputusan kelulusan.

Permohonan Anda Ditolak
Sila hubungi pihak kami untuk maklumat lanjut.

Permohonan Anda Dalam Penilaian
Pihak kami sedang menilai permohonan anda. Sila semak semula kemudian.

Figure 48: Check Loan Application Interface

10.2.8 Check Member Saving

Koperasi Kakitangan KADA

Pilihan

DAFTAR KELUAR

SIMPANAN DAN SYER PERIBADI

Nama: Harini A/P. Sangaran

No. IC: 041201070330

No. Ahli: 0081

Simpanan Peribadi Ahli

Nombor akaun	Baki Simpanan (RM)	Tarikh Kemaskini
1	100.00	2025-01-08 02:26:54
1	90.00	2024-11-20 02:27:32
2	50.00	2023-12-19 02:28:26
2	130.00	2025-01-14 02:28:47
2	230.00	2024-12-26 02:33:06

Syer Ahli

Syer Majikan	Syer Pekerja	Tarikh
70.00	55.00	2025-01-08 02:49:56
85.00	70.00	2024-11-20 02:50:16
60.00	40.00	2023-12-19 02:50:29
90.00	65.00	2025-01-14 02:50:40
180.00	110.00	2024-12-26 02:50:52

Figure 49: Check Member Saving Interface

10.2.9 Check Loan Balance

Koperasi Kakitangan KADA

Pilihan DAFTAR KELUAR

BAKI PINJAMAN

RM 2593.50

Jenis Pinjaman: Baik Pulih Kenderaan
Ansuran Bulanan: RM 135.50

Tarikh	Bayaran	No. Rujukan	Kaedah Pembayaran
2024-11-14	RM 135.50	P001	FPX Method
2024-12-19	RM 135.50	P002	FPX Method
2025-01-09	RM 135.50	P003	FPX Method

Figure 50: Check Loan Balance Interface

10.2.10 Check Invoice/Transaction History

Koperasi Kakitangan KADA

Pilihan DAFTAR KELUAR

PENYATA KEWANGAN

Nama: Harini A/P Sangaran
No. IC: 041201070330
No. Ahli: 0081

Transaksi	Tarikh	Jumlah (RM)	Catatan	Cara Bayaran
1	2024-11-14	135.50	P001	FPX Method
2	2024-12-19	135.50	P002	FPX Method
3	2025-01-09	135.50	P003	FPX Method

Figure 51: Check Invoice/Transaction History Interface

10.2.11 Apply Membership Termination Form

Koperasi Kakitangan KADA Pilihan DAFTAR KELUAR

Borang Permohonan Berhenti Keahlian

Nama	No KP	
Tarikh Lahir dd/mm/yyyy	Umur	
Alamat Rumah		
Poskod	Bandar	Negeri
No Ahli	No PF	
Jawatan	Alamat Pejabat	
Poskod Pejabat	Bandar Pejabat	No Tel/Fax

No Tel Bimbit	Jantina Lelaki
Agama Islam	Nyatakan (Jika Lain-lain)
Bangsa Melayu	Nyatakan (Jika Lain-lain)
Sebab Berhenti	

Hantar Permohonan

Figure 52: Apply Membership Termination Interface

10.2.12 Navigation Bar



Figure 53: Navigation Bar Interface

10.3 Admin User

10.3.1 Admin Login

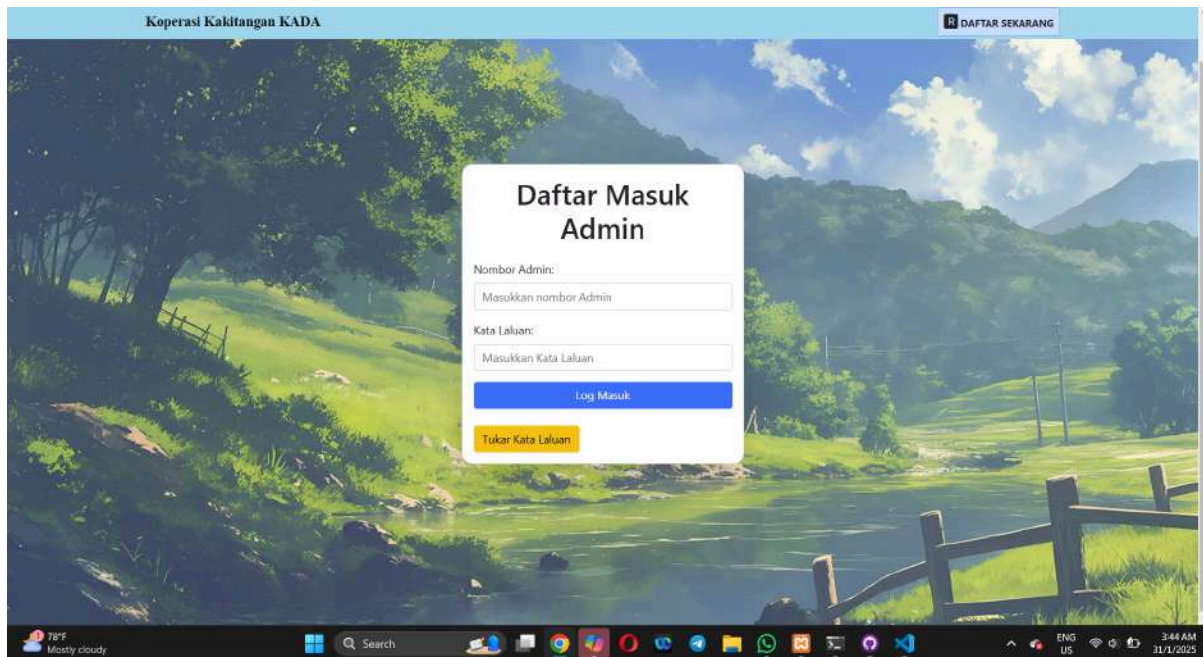


Figure 54: Admin Login Interface

10.3.2 Admin Homepage

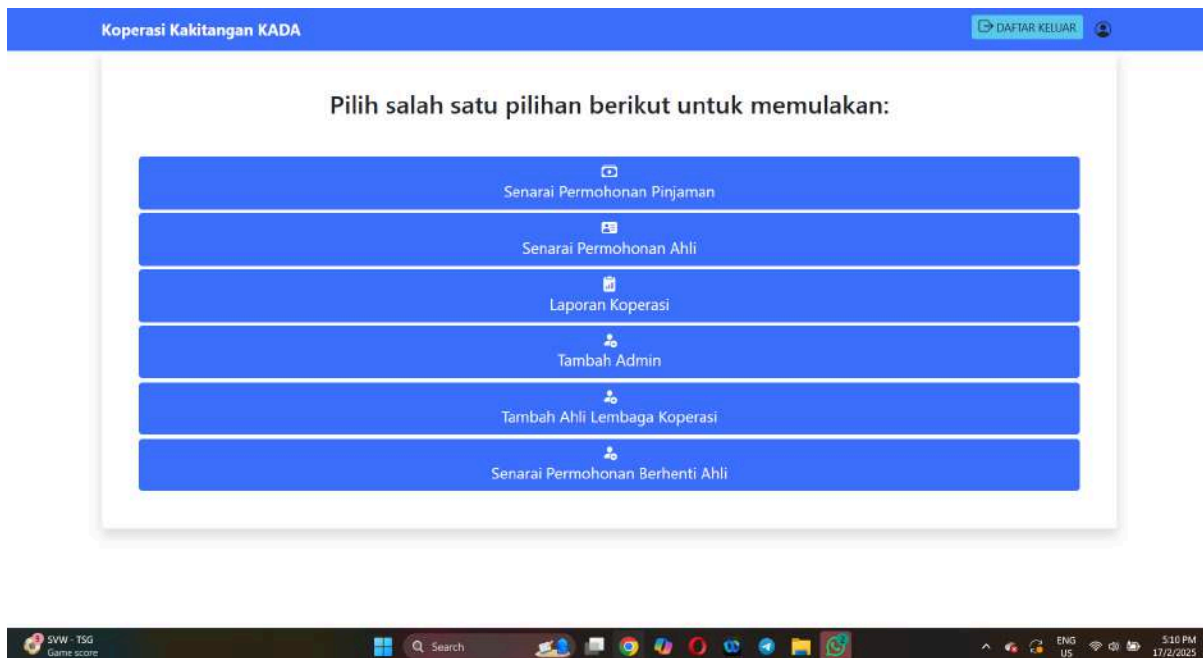


Figure 55: Admin Homepage Interface

10.3.3 View Membership Application List

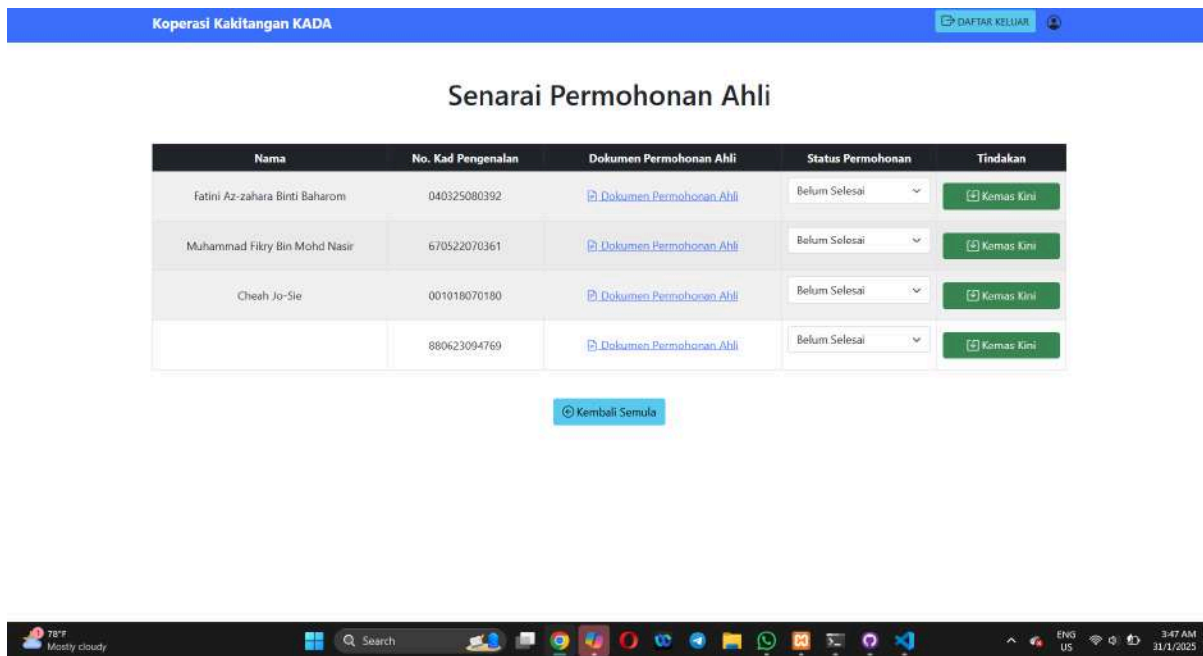


Figure 56: View Membership Application List Interface

10.3.4 Successful Membership Application Status Update

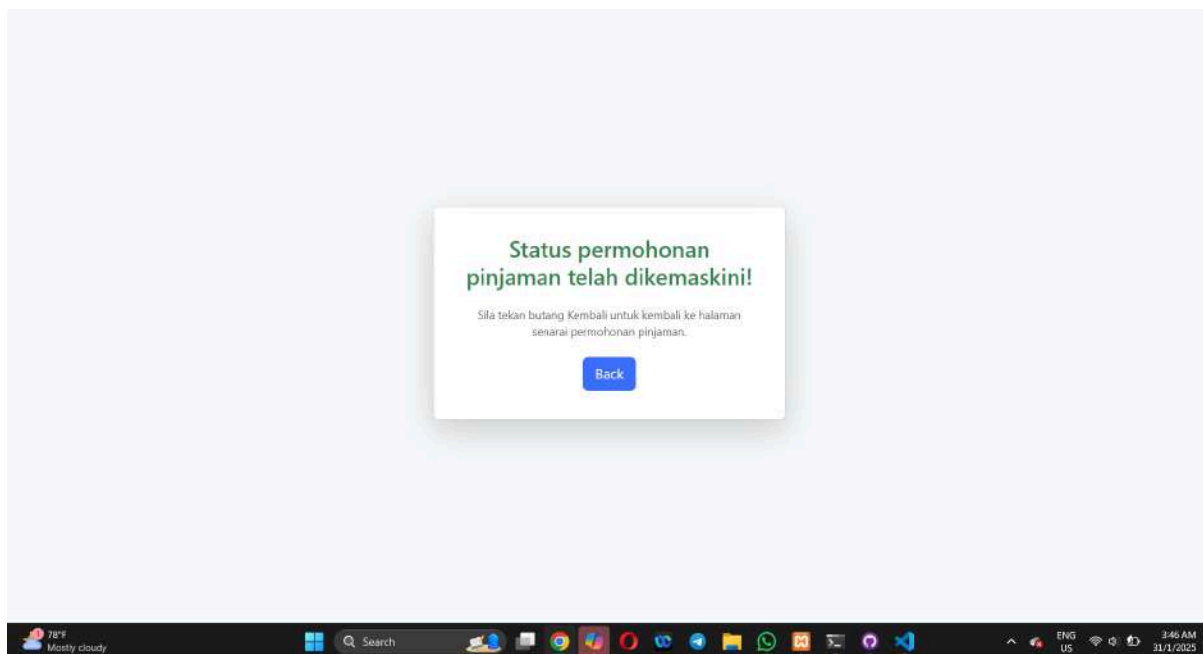


Figure 57: Successful Membership Application Status Update Interface

10.3.5 View Loan Application List

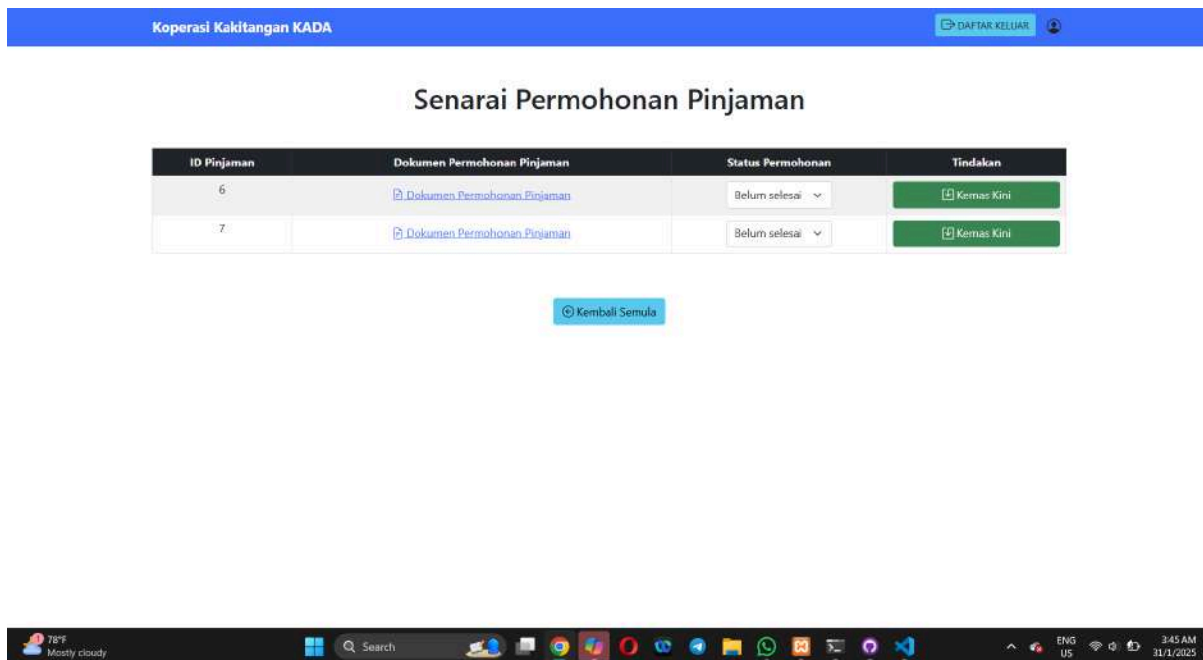


Figure 58: View Loan Application List Interface

10.3.6 Successful Loan Application Status Update

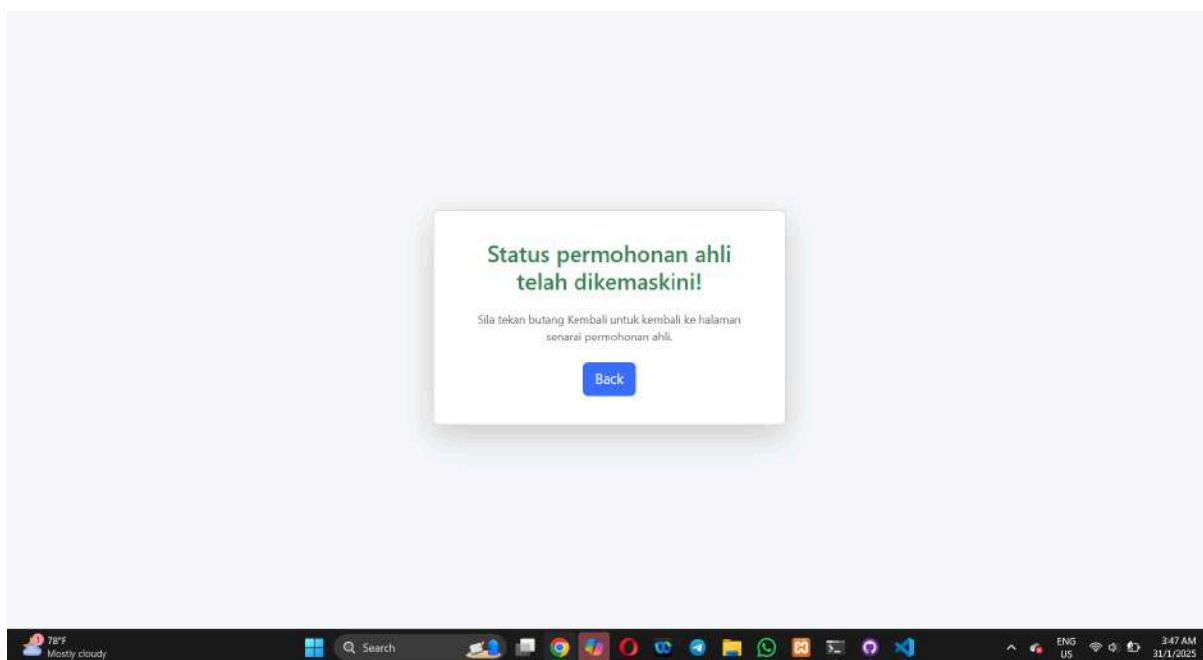


Figure 59: Successful Loan Application Status Update Interface

10.3.7 View Monthly and Annual Report

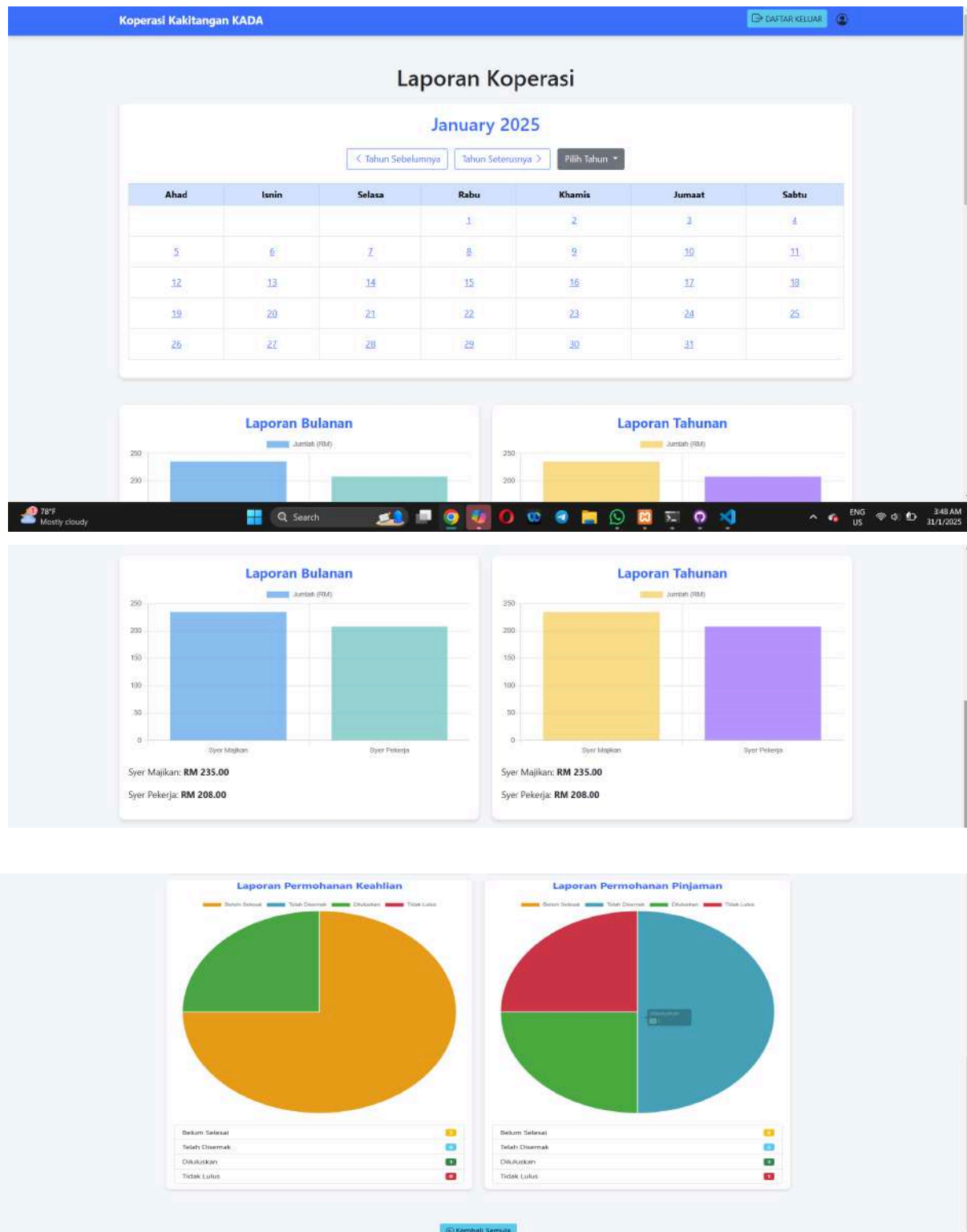


Figure 60: Monthly and Annual Report Interface

10.3.8 Add New Admin

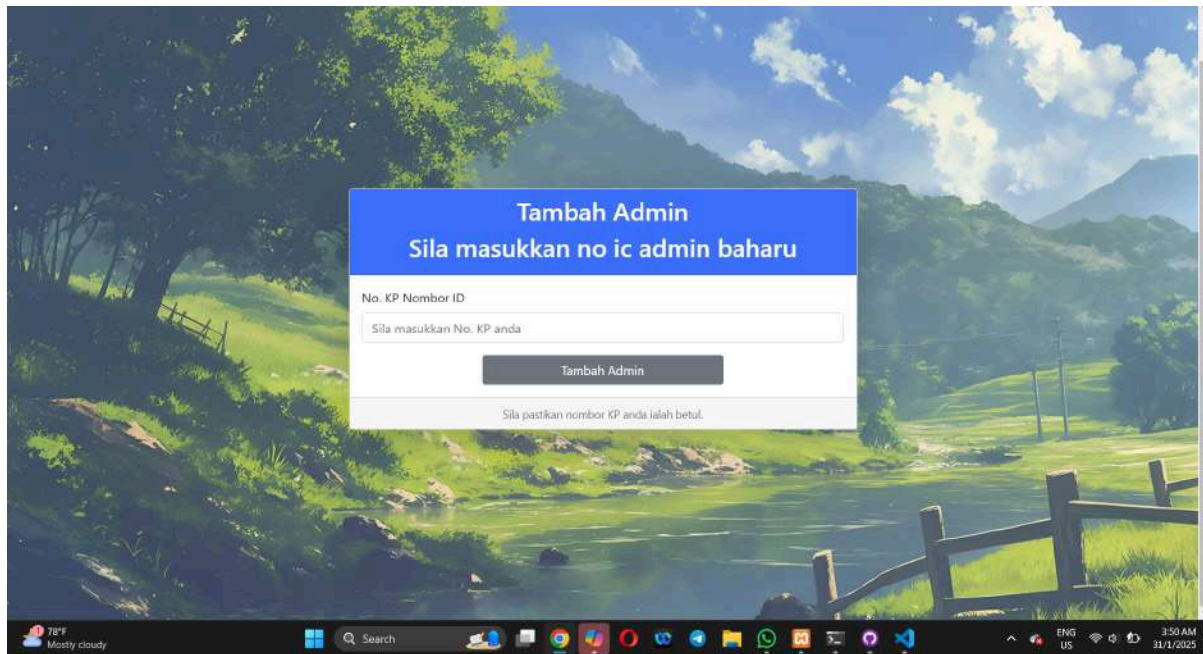


Figure 61: Add New Admin Interface

10.3.9 Add New ALK

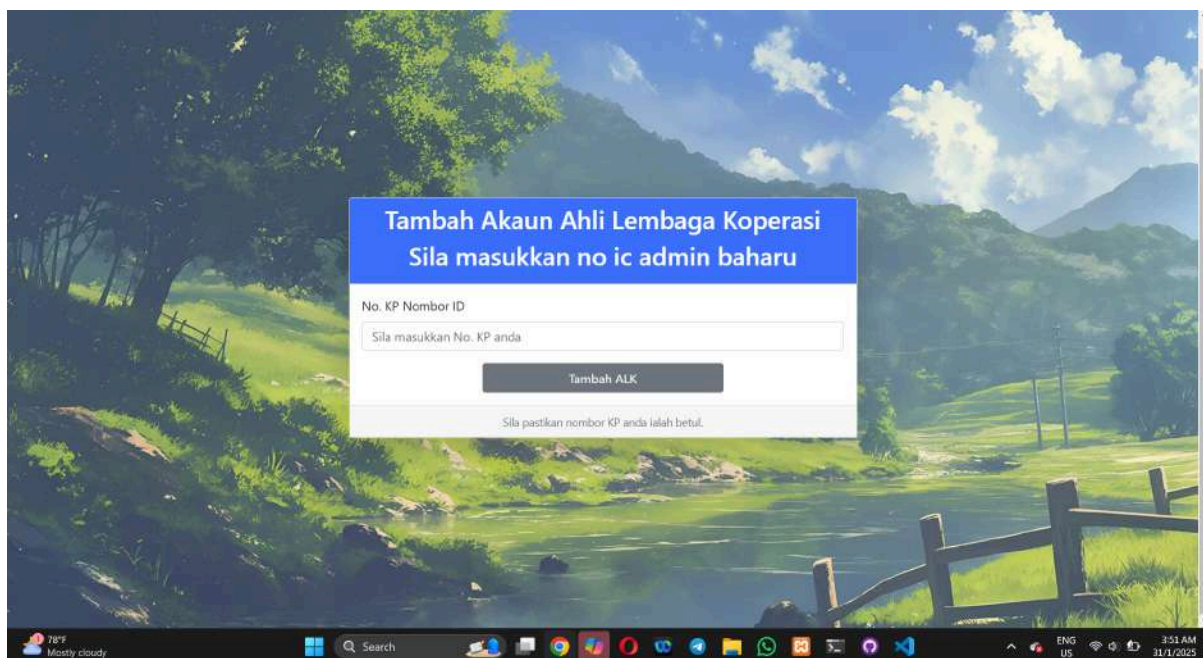


Figure 62: Add New ALK Interface

10.3.10 Successful Add New Admin/ALK

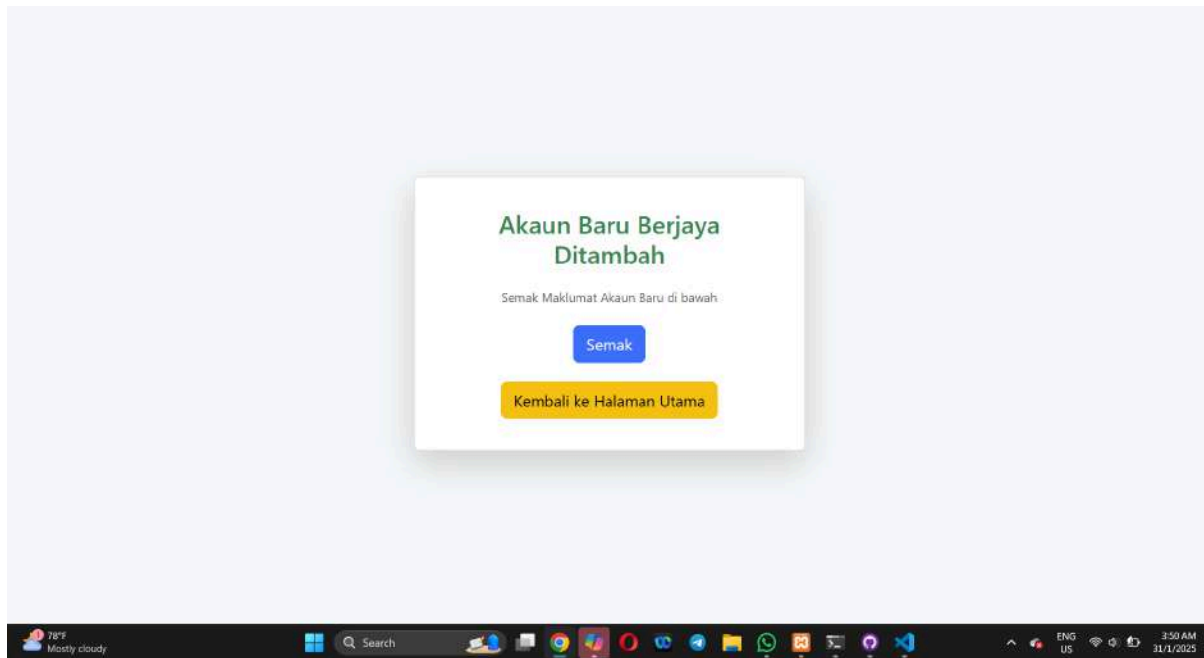


Figure 63: Successful Add New Admin/ALK Interface

10.3.11 View Membership Discontinuation List

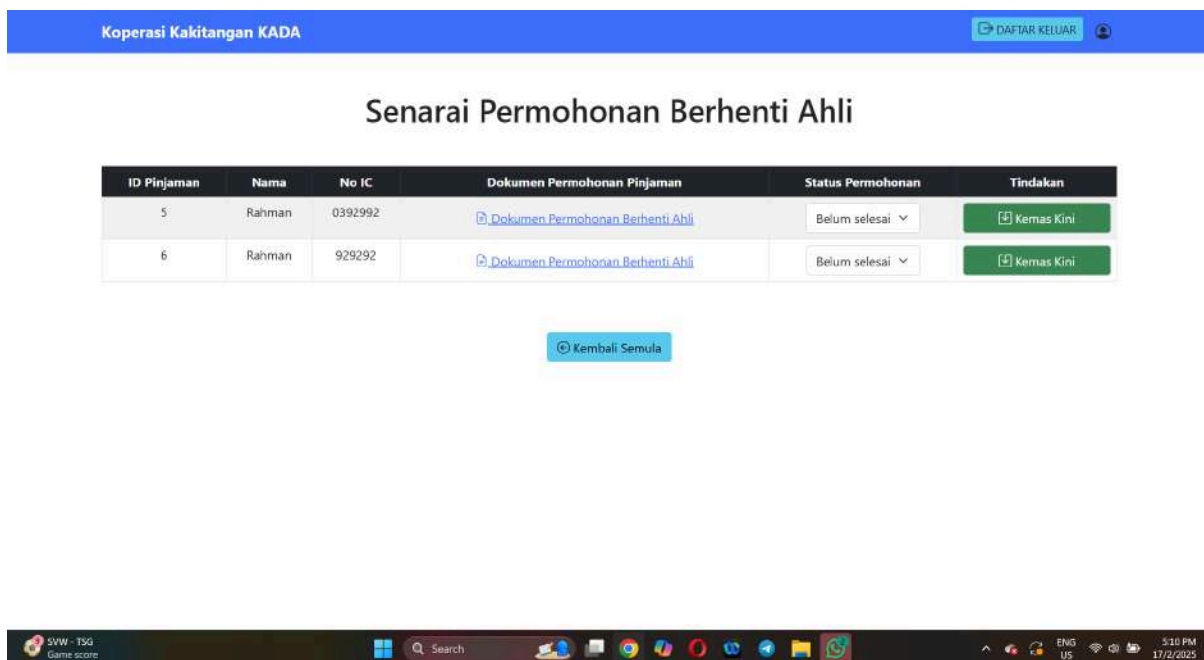


Figure 64: View Membership Discontinuation List Interface

10.4 ALK User

10.4.1 ALK Login

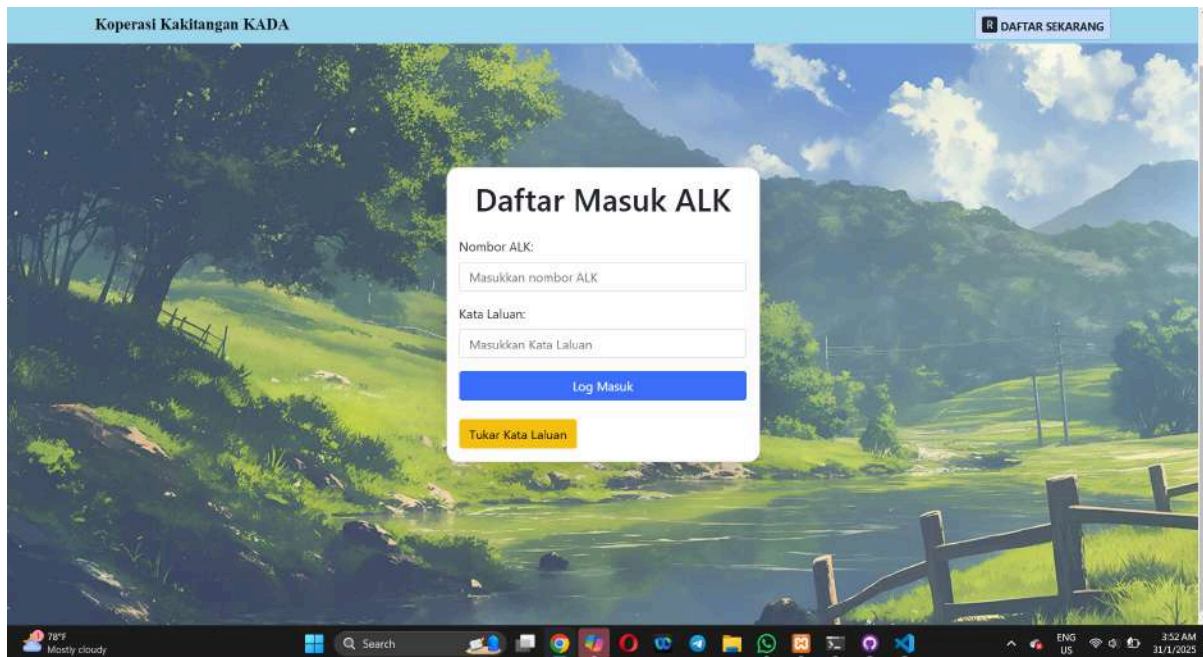


Figure 65: ALK Login Interface

10.4.2 ALK Homepage

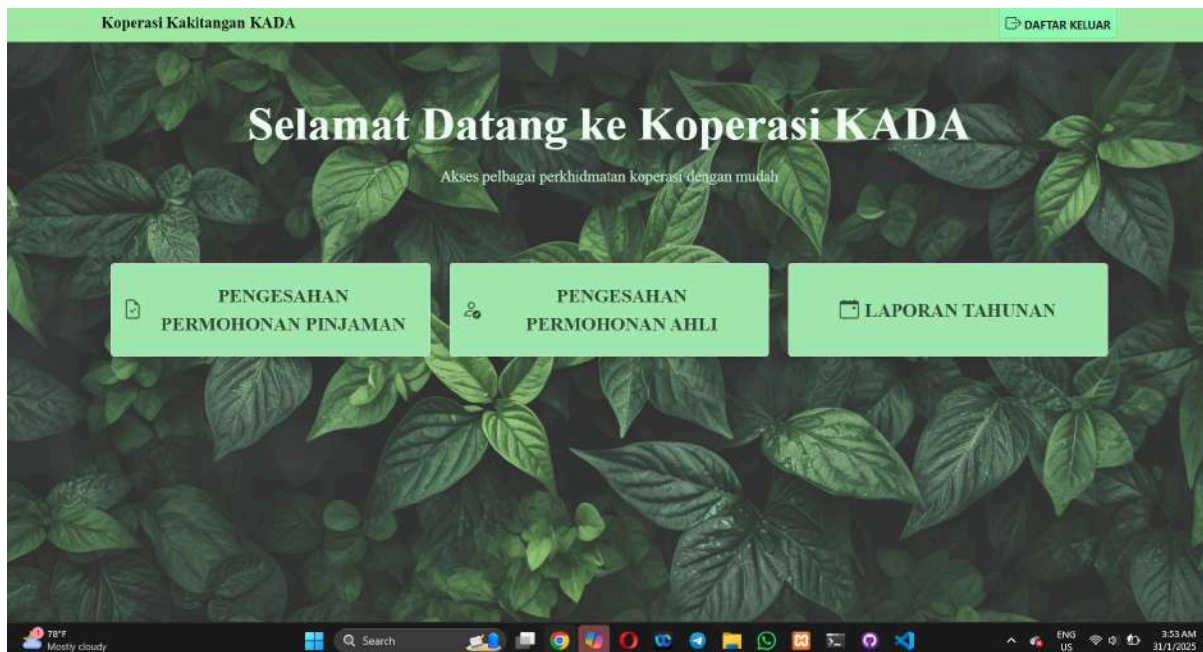


Figure 66: ALK Homepage Interface

10.4.3 Approve Loan Application

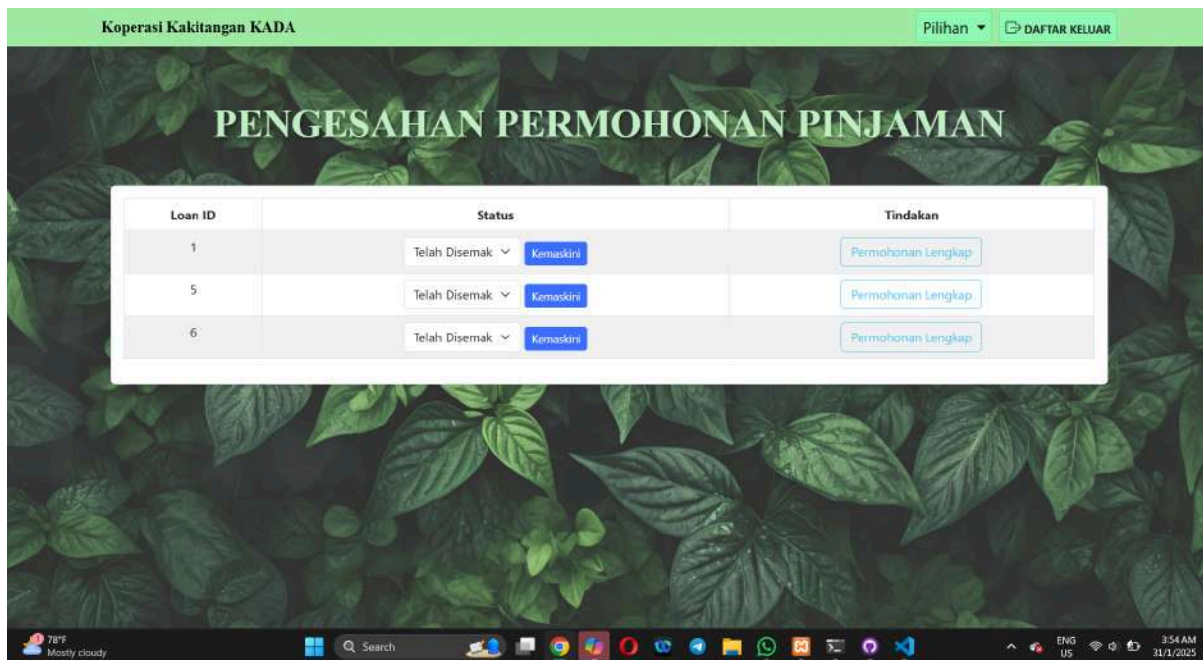


Figure 67: Approve Loan Application Interface

10.4.4 Successful Loan Application Status Update

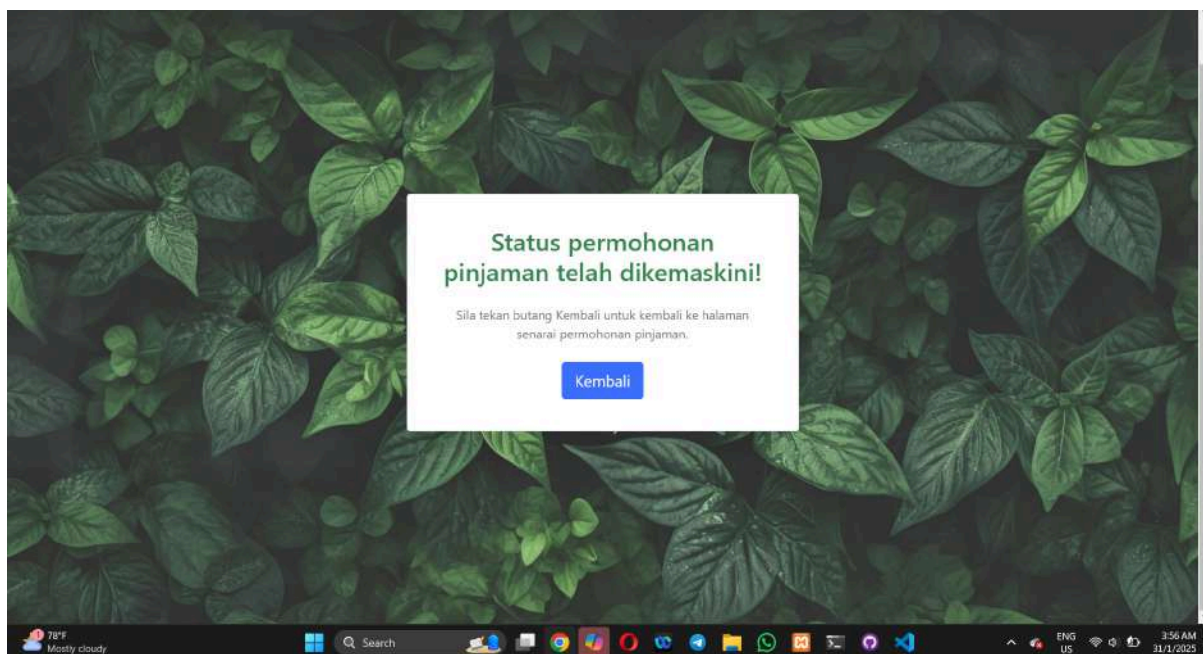


Figure 68: Successful Loan Application Status Update Interface

10.4.5 Approve Membership Application

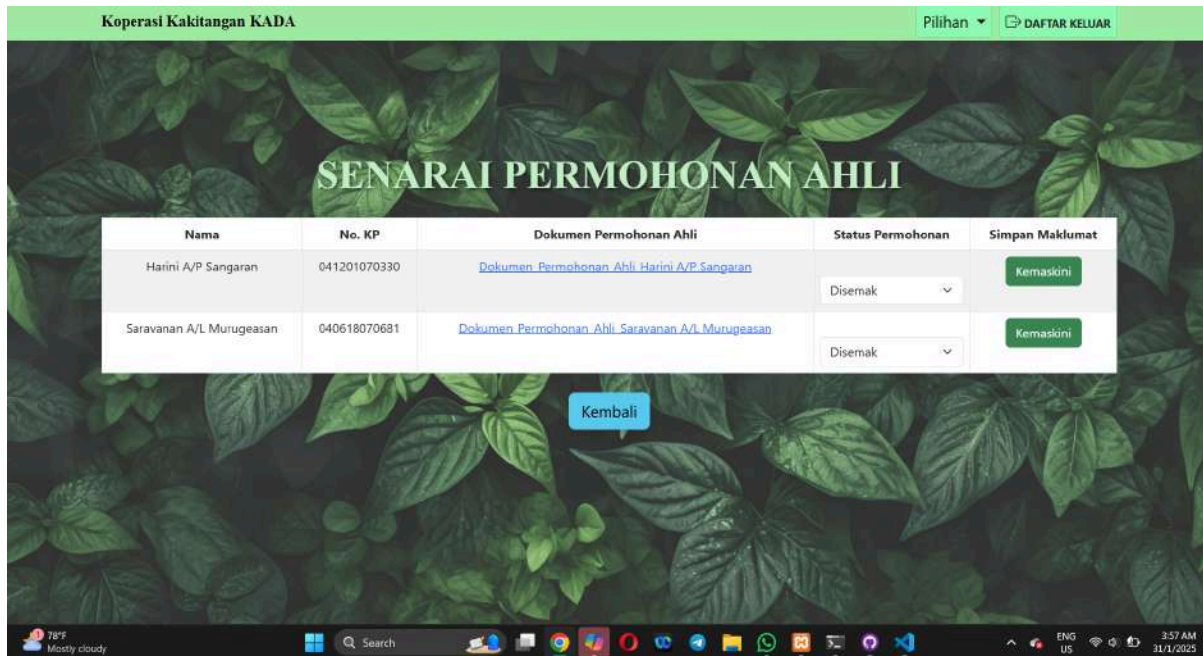


Figure 69: Approved Membership Application Interface

10.4.6 Successful Membership Application Status

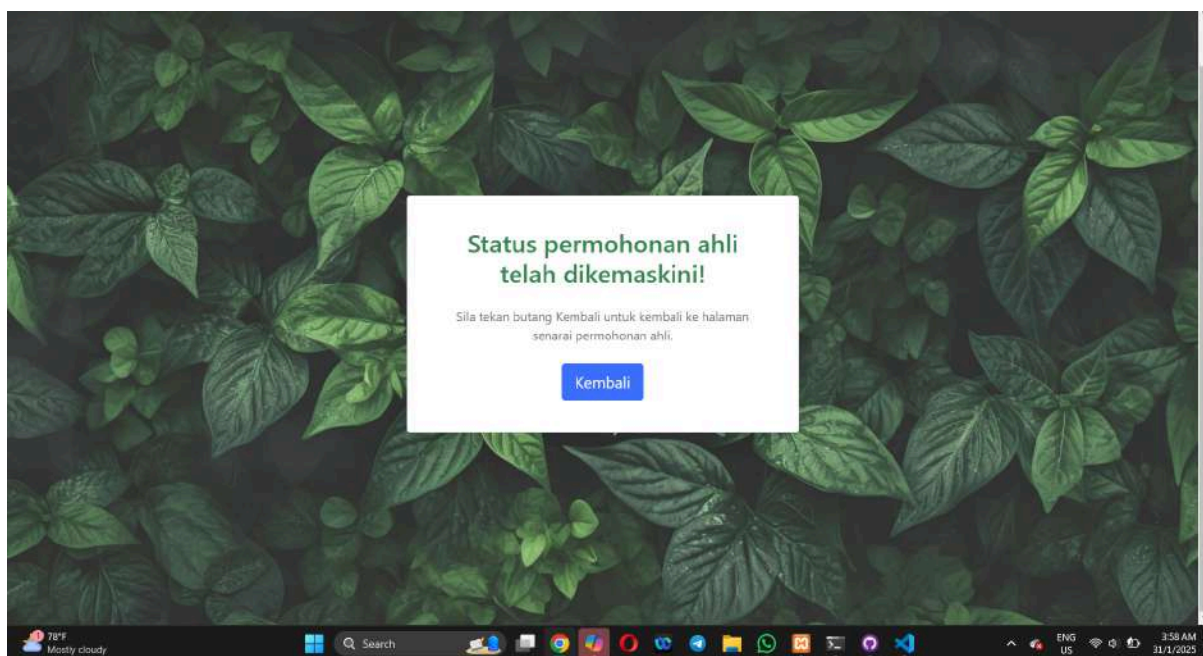


Figure 70: Successful Membership Application Status Update Interface

10.4.7 Navigation Bar

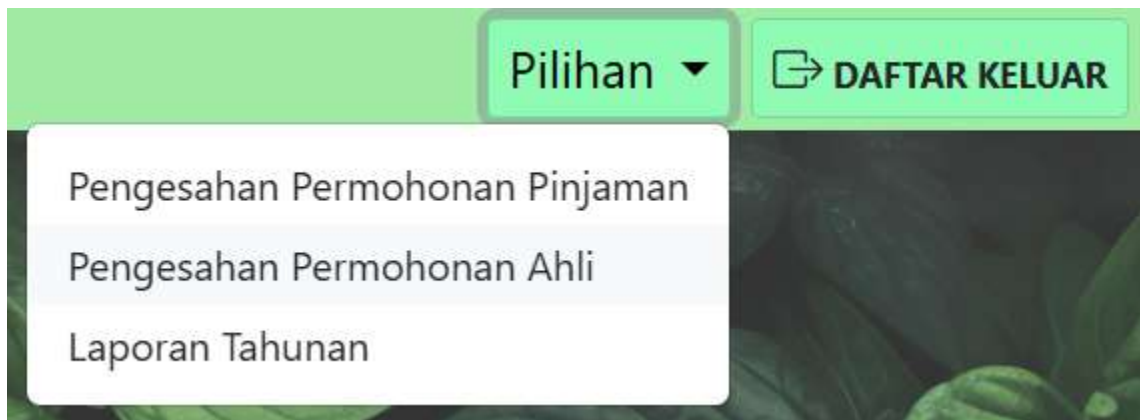
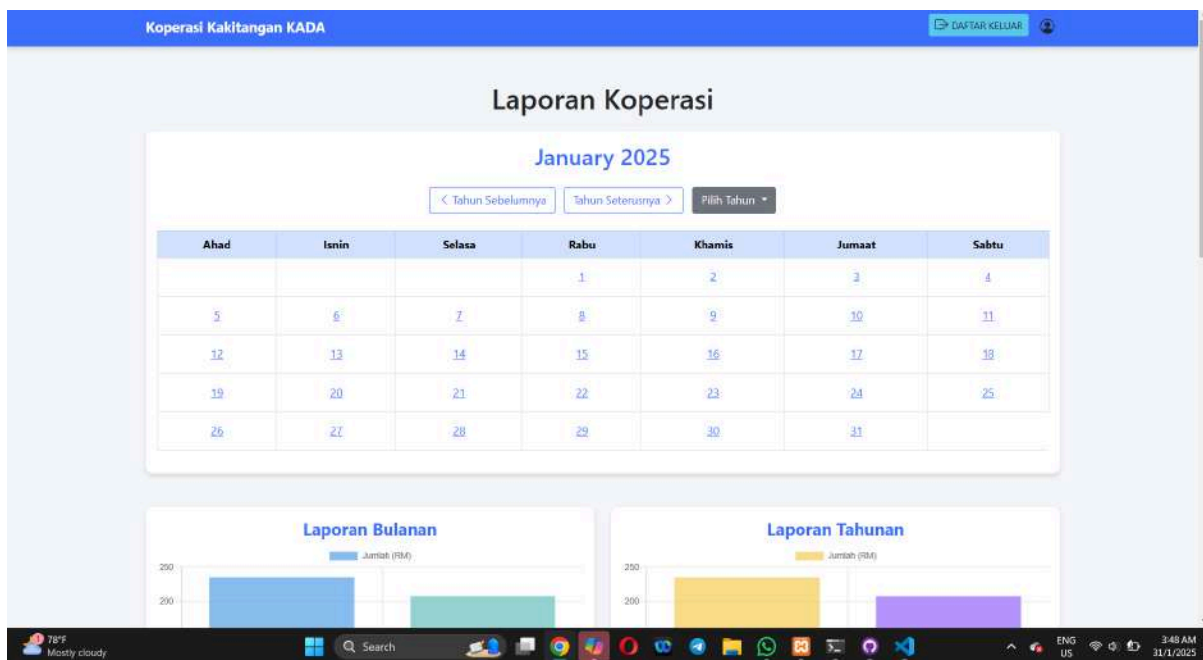


Figure 71: Navigation Bar Interface

10.3.7 View Monthly and Annual Report



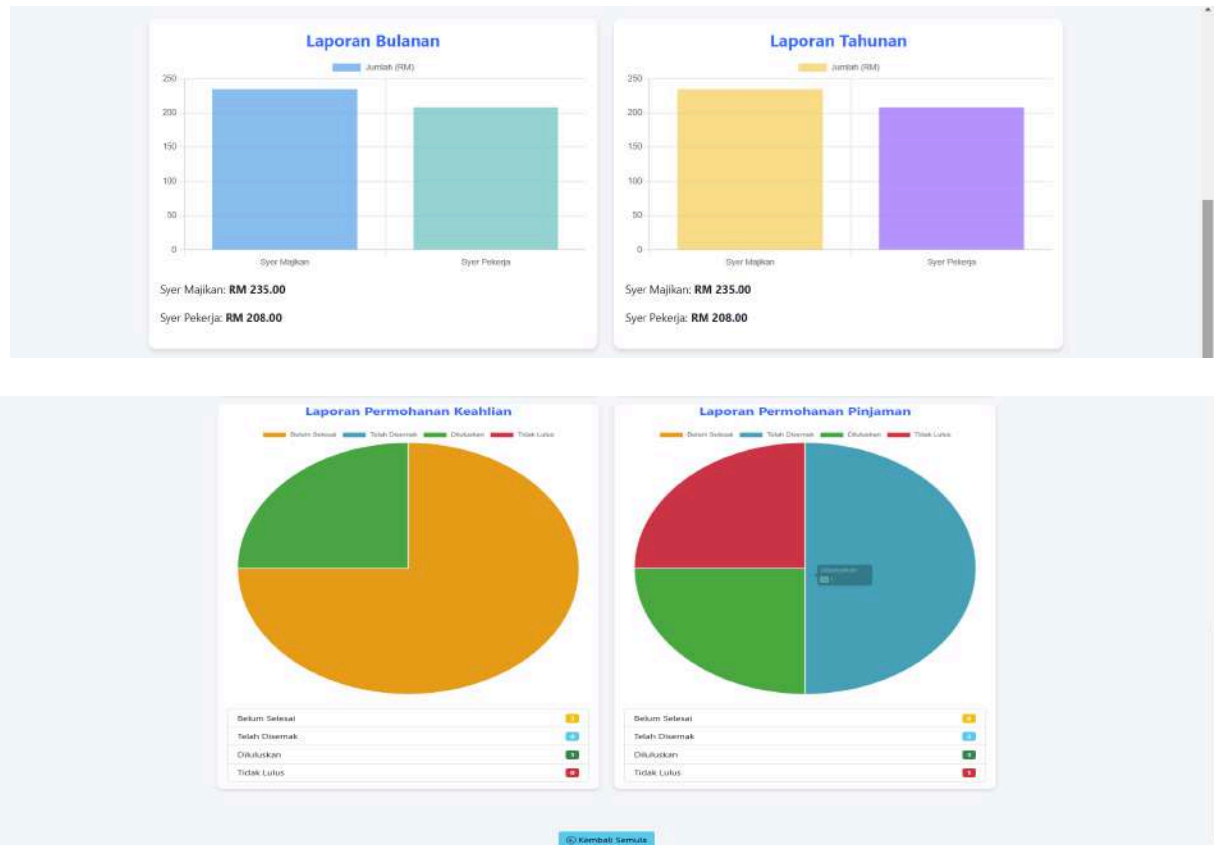


Figure 72: Monthly and Annual Report Interface

10.5 All User

10.5.1 Reset Password



The screenshot shows a web application window with a title bar. The main content area has a blue header with the text "Tukar Kata Laluan". Below the header, there are two input fields: "No. KP" and "Kata Laluan Baharu". A blue button labeled "Simpan Maklumat" is positioned at the bottom right of the form area. The Windows taskbar is visible at the bottom of the screen, showing the time as 3:23 AM on 31/1/2025.

Figure 73: Reset Password Interface

10.5.2 Successful Reset Password



The screenshot shows a web application window with a title bar. The main content area has a blue header with the text "Kemaskini Kata Laluan". Below the header, there is a green message box with the text "Kata laluan anda telah berjaya dikemaskini!". At the bottom, there are two buttons: "Kembali ke Laman Utama" (grey) and "Tukar Kata Laluan Lagi" (yellow). The Windows taskbar is visible at the bottom of the screen, showing the time as 3:24 AM on 31/1/2025.

Figure 74: Successful Reset Password Interface

10.5.3 Unsuccessful Reset Password

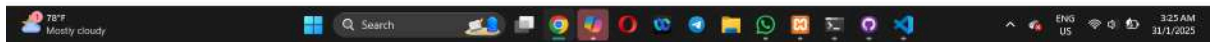


Figure 75: Unsuccessful Reset Password Interface

10.5.4 Log Out



Figure 76: Log Out Interface

11.0 Alpha and Beta Testing Reports

In this section, we are going to explain the tests that are done to our system.

11.1 Alpha Testing Report

11.1.0 Equivalence Partitioning and Boundary Value Analysis

Object 1: Non-member

Object Class	Attributes	Equivalence Classes	Boundary Values
Member Application	applicant_id	Valid: 1, 100, 1000 Invalid: -1, 1.10, abc	Minimum: 1 Maximum: 2,147,483,647
	name	Valid: "John Doe", "Ali Ahmad" Invalid: "", "J"	Minimum length: 2 characters Maximum length: 255 characters
	ic_number	Valid: "900101-14-5678" Invalid: "1234", "abc"	Minimum length: 12 characters Maximum length: 20 characters
	email	Valid: "user@example.com", "info@domain.net" Invalid: "user123.com", "test@domain"	Minimum length: a@domain.com Maximum length: 255 characters
	marital_status	Valid: "Bujang", "Berkahwin", "Duda", "Janda" Invalid: "Single", "Married"	Enum Values: 4 fixed values
	gender	Valid: "Lelaki", "Perempuan" Invalid: "Male", "Female"	Enum Values: 2 fixed values
	age	Valid: 18, 30, 60 Invalid: 0, -5, 150	Minimum: 18 Maximum: 100
	religion	Valid: "Islam", "Christian", "Hindu" Invalid: "None", "XYZ"	Minimum length: 3 characters Maximum length: 50 characters
	ethnicity	Valid: "Malay", "Chinese",	Minimum length: 3

	"Indian" Invalid: "123", "Unknown"	characters Maximum length: 50 characters
home_address	Valid: "No 10, Jalan ABC, Kuala Lumpur" Invalid: ""	Minimum length: 10 characters Maximum length: 500 characters
postcode	Valid: "50000", "12345" Invalid: "500", "abc12"	Minimum length: 5 characters Maximum length: 10 characters
position_grade	Valid: "Gred 41", "Gred 44" Invalid: "Level 1", "Grade A"	Minimum length: 3 characters Maximum length: 100 characters
phone_mobile	Valid: "012-3456789", "+6012-3456789" Invalid: "12345", "phone123"	Minimum length: 10 characters Maximum length: 15 characters
monthly_salary	Valid: 3000, 10000 Invalid: -500, "abc"	Minimum: 1000 Maximum: 1,000,000
famName1	Valid: "Ahmad Ali", "Siti Nor" Invalid: ""	Minimum length: 2 characters Maximum length: 255 characters
famRelationship1	Valid: "Bapa", "Ibu" Invalid: ""	Minimum length: 2 characters Maximum length: 255 characters
famiC1	Valid: "900101-14-5678" Invalid: "1234", "abc"	Minimum length: 12 characters Maximum length: 20 characters
famName2	Valid: "Ahmad Ali", "Siti Nor" Invalid: ""	Minimum length: 2 characters Maximum length: 255 characters

famRelationship2	Valid: "Bapa", "Ibu" Invalid: ""	Minimum length: 2 characters Maximum length: 255 characters
famiC2	Valid: "900101-14-5678" Invalid: "1234", "abc"	Minimum length: 12 characters Maximum length: 20 characters
yuran_masuk_value	Valid: 100.00, 5000.50 Invalid: -100, "abc"	Minimum: 0.01 Maximum: 1,000,000.00
modal_syer_value	Valid: 100.00, 5000.50 Invalid: -100, "abc"	Minimum: 0.01 Maximum: 1,000,000.00
modal_yuran_value	Valid: 100.00, 5000.50 Invalid: -100, "abc"	Minimum: 0.01 Maximum: 1,000,000.00
wang_deposit_value	Valid: 100.00, 5000.50 Invalid: -100, "abc"	Minimum: 0.01 Maximum: 1,000,000.00
sumbangan_kebajikan_value	Valid: 100.00, 5000.50 Invalid: -100, "abc"	Minimum: 0.01 Maximum: 1,000,000.00
simpanan_tetap_value	Valid: 100.00, 5000.50 Invalid: -100, "abc"	Minimum: 0.01 Maximum: 1,000,000.00
lain_lain_value	Valid: 100.00, 5000.50 Invalid: -100, "abc"	Minimum: 0.01 Maximum: 1,000,000.00
approval	Valid: "Pending", "Reviewed", "Disapproved", "Approved" Invalid: "Yes", "No"	Enum Values: 4 fixed values

Table 1: Equivalence and Boundary Analysis for Membership Application

Object 2: Member:

Object Class	Attributes	Equivalence Classes	Boundary Values
Member Termination Request	id	Valid: 1, 100, 1000 Invalid: -1, 1.10, abc	Minimum: 1 Maximum: 2,147,483,647
	member_id	Valid: 0001, 0012, 1234 Invalid: "abc", -5	Minimum: 0001 Maximum: 9999
	name	Valid: "John Doe", "Ali Ahmad" Invalid: "", "J"	Minimum length: 2 characters Maximum length: 255 characters
	no_kp	Valid: "900101-14-5678" Invalid: "1234", "abc"	Minimum length: 12 characters Maximum length: 20 characters
	tarikh_lahir	Valid: "1990-05-20", "1985-12-01" Invalid: "2025-10-01", "abcd"	Date Format: YYYY-MM-DD
	umur	Valid: 18, 45, 60 Invalid: -5, 150	Minimum: 18 Maximum: 100
	gender	Valid: "Lelaki", "Perempuan" Invalid: "Male", "Female"	Enum Values: 2 fixed values
	age	Valid: 18, 30, 60 Invalid: 0, -5, 150	Minimum: 18 Maximum: 100
	tarikh_permohonan	Valid: "2024-01-10", "2023-12-31" Invalid: "abcd"	Date Format: YYYY-MM-DD
	status	Valid: "Pending", "Approved", "Rejected" Invalid: "Yes", "No"	Enum Values: 3 fixed values

Table 2: Equivalence and Boundary Analysis for Member Termination Request

Object Class	Attributes	Equivalence Classes	Boundary Values
Loan Application	LoanID	Valid: 1, 100, 1000 Invalid: -1, 1.10, abc	Minimum: 1 Maximum: 2,147,483,647
	member_id	Valid: 0001, 0012, 1234 Invalid: "abc", -5	Minimum: 0001 Maximum: 9999
	LoanType	Valid: "Al-Bai", "Al-Inah", "Skim Khas" Invalid: "Personal Loan", "House Loan"	Enum Values: 6 fixed values
	LoanAmount	Valid: 1000.00, 50000.00 Invalid: -500, "abc"	Minimum: 1000.00 Maximum: 1,000,000.00
	RepaymentPeriodMonths	Valid: 12, 24, 60 Invalid: -6, 0, "abc"	Minimum: 6 months Maximum: 360 months
	MonthlyInstallment	Valid: 100.00, 5000.00 Invalid: -10, "xyz"	Minimum: 50.00 Maximum: 50,000.00
	approval	Valid: "Pending", "Reviewed", "Disapproved", "Approved" Invalid: "Yes", "No"	Enum Values: 4 fixed values
	BankName	Valid: "Maybank", "CIMB" Invalid: "", "XYZ Bank 123"	Minimum length: 2 Maximum length: 100
	AccountNumber	Valid: "123456789012" Invalid: "1234", "abcdef"	Minimum length: 10 Maximum length: 20

Table 3: Equivalence and Boundary Analysis for Loan Application

Object 3: Admin

Object Class	Attributes	Equivalence Classes	Boundary Values
Add ALK	ALK_id	Valid: 1, 100, 1000 Invalid: -1, 1.10, abc	Minimum: 1 Maximum: 2,147,483,647
	Password	Valid: "P@ssword123", "MySecurePass99!" Invalid: "", "123"	Minimum length: 8 Maximum length: 255
	no_kp	Valid: "900101-14-5678" Invalid: "1234", "abc"	Minimum length: 12 characters Maximum length: 20 characters
	user_role	Valid: "ALK" Invalid: "Admin", "User", "123"	Enum Values: Fixed value "ALK"

Table 4: Equivalence and Boundary Analysis for Add ALK

Object Class	Attributes	Equivalence Classes	Boundary Values
Add Admin	Admin_id	Valid: 1, 100, 1000 Invalid: -1, 1.10, abc	Minimum: 1 Maximum: 2,147,483,647
	Password	Valid: "P@ssword123", "MySecurePass99!" Invalid: "", "123"	Minimum length: 8 Maximum length: 255
	no_kp	Valid: "900101-14-5678" Invalid: "1234", "abc"	Minimum length: 12 characters Maximum length: 20 characters
	user_role	Valid: "ADMIN" Invalid: "ALK", "User", "123"	Enum Values: Fixed value "ADMIN"

Table 5: Equivalence and Boundary Analysis for Add Admin

11.1.1 Test Cases Based on Subsystem

TC001: Test Member Subsystem: <User Story 1 (US001)>

This test contains the following test cases:

- (a) TC001_01: Test <View Member Menu (SD024)>
- (b) TC001_02: Test <Check Current Saving (SD003)>
- (c) TC001_03: Test <Apply Membership Termination (SD006)>

DESCRIPTION	VERSION	TEST DATE
View Member Menu	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the members are able to view the appropriate homepage once they have successfully logged in.	Damiya Aina binti Basir Abd Shammad	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Harini A/P Sangaran	-
2	View Member homepage.	-	Display the features for member to do in the system	As Expected	Pass	Harini A/P Sangaran	

Table 6: TC001_01: Test <View Member Menu (SD024)>

DESCRIPTION	VERSION	TEST DATE
Check Current Saving	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
Ensure that the system correctly displays the member's savings including the personal and member syer	Nuraisyah binti Mohd Zikre	Damiya Aina binti Basir Abd Shammad

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Damiya Aina binti Basir Abd Shammd	-
2	Navigate to the Semak Simpanan section from the Member homepage.	-	Successfully load the member saving section, displaying all the information of the member saving including the personal saving and member syer	As Expected	Pass	Damiya Aina binti Basir Abd Shammd	
3	view the table saving and member syer	-	table saving and member syer is successfully appear	As Expected	Pass	Damiya Aina binti Basir Abd Shammd	

Table 7: TC001_02: Test <Check Current Saving (SD003)>

DESCRIPTION	VERSION	TEST DATE
Applying the Membership Termination Application	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
allows non-members to apply for membership termination by completing the form and giving the reasonable reason of quitting	Nurul Ika Syafiny binti Azhar	Damiya Aina binti Basir Abd Shammad

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
2	Navigate to Mohon Berhenti Ahli section from the member homepage	-	Successfully load the membership termination section, displaying all the form that need to be filled in	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
3	Fill in the membership termination application form fields (e.g: nama, no.kp, etc)	nama: Ali bin Abu no.kp: 780506030201	Form is completed	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	
4	click Hantar Permohonan button after completing the form	-	successfully submit the membership termination application form	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	

Table 8: TC001_03: Test <Apply Membership Termination (SD006)>

TC002: Test Admin Subsystem: <User Story 2 (US002)>

This test contains the following test cases:

- (a) TC002_01: Test <View Staff Menu (SD025)>
- (b) TC002_02: Test <View Member Application Form (SD007)>
- (c) TC002_03: Test <View Loan Application Form (SD008)>
- (d) TC002_04: Test <View Membership Termination Form (SD009)>
- (e) TC002_05: Test <Validate Member Application Form (SD010)>
- (f) TC002_06: Test <Validate Loan Application Form (SD011)>
- (g) TC002_07: Test <Approve Membership Termination (SD012)>
- (f) TC002_08: Test <View Monthly Report (SD013)>
- (g) TC002_09: Test <View Annual Report (SD014)>
- (h) TC002_10: Test <Add New Admin (SD015)>
- (i) TC002_11: Test <Add New ALK (SD016)>

DESCRIPTION	VERSION	TEST DATE
View Staff Menu	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admins are able to view the appropriate homepage once they have successfully logged in.	Nurul Ika Syafiny binti Azhar	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Harini A/P Sangaran	-
2	View admin homepage.	-	Display the features for admin to do in the system	As Expected	Pass	Harini A/P Sangaran	-

Table 9: TC002_01: Test <View Staff Menu (SD017)>

DESCRIPTION	VERSION	TEST DATE
View Member Application Form	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admins are able to view the list of pending member application forms.	Harini A/P Sangaran	Damiya Aina binti Basir Abd Shammad

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	View admin homepage.	-	Display the features for admin to do in the system	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-
2	Click on the 'Senarai Permohonan Ahli' button.	-	Display list of pending member application forms.	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-

Table 10: TC002_02: Test <View Member Application Form (SD007)>

DESCRIPTION	VERSION	TEST DATE
View Loan Application Form	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admins are able to view the list of pending loan application forms.	Damiya Aina binti Basir Abd Shammad	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	View admin homepage.	-	Display the features for admin to do in the system	As Expected	Pass	Nuraisyah binti Mohd Zikre	-
2	Click on the 'Senarai Permohonan Pinjaman' button.	-	Display list of pending loan application forms.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-

Table 11: TC002_03: Test <View Loan Application Form (SD008)>

DESCRIPTION	VERSION	TEST DATE
View Membership Termination Form	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admins are able to view the list of pending membership termination forms.	Nurul Ika Syafiny binti Azhar	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	View admin homepage.	-	Display the features for admin to do in the system	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-
2	Click on the 'Senarai Permohonan Berhenti Ahli' button.	-	Display list of pending membership termination forms.	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-

Table 12: TC002_04: Test <View Membership Termination Form (SD009)>

DESCRIPTION	VERSION	TEST DATE
Validate Member Application Form	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admins are able to view applicant's verification documents and update the application status.	Nuraisyah binti Mohd Zikre	Harini A/P Sangaran

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Senarai Permohonan Ahli' button.	-	Display list of pending member application forms.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
2	Click on the 'Dokumen Permohonan Ahli' hyperlink.	-	Display applicant's application form.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
3	Click on the drop down button in the 'Status Permohonan' column.	-	Display the current application status by default and display all the possible statuses when clicked on the drop down button.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
4	Click on the 'Kemas Kini' button.	-	Update applicant's application status.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-

Table 13: TC002_05: Test <Validate Member Application Form (SD010)>

DESCRIPTION	VERSION	TEST DATE
Validate Loan Application Form	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admins are able to view member's verification documents and update the loan application status.	Nurul Ika Syafiny binti Azhar	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Senarai Permohonan Pinjaman' button.	-	Display list of pending loan application forms.	As Expected	Pass	Harini A/P Sangaran	-
2	Click on the 'Dokumen Permohonan Pinjaman' hyperlink.	-	Display member's loan application form.	As Expected	Pass	Harini A/P Sangaran	-
3	Click on the 'Muat Turun Dokumen' button.	-	Download member's verification documents for the loan application	As Expected	Pass	Harini A/P Sangaran	-
4	Click on the drop down button in the 'Status Permohonan' column.	-	Display the current application status by default and display all the possible statuses when clicked on the drop down button.	As Expected	Pass	Harini A/P Sangaran	-
5	Click on the 'Kemas Kini' button.	-	Update member's loan application status.	As Expected	Pass	Harini A/P Sangaran	-

Table 14: TC002_05: Test <Validate Loan Application Form (SD011)>

DESCRIPTION	VERSION	TEST DATE
Approve Membership Termination Form	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admins are able to view membership termination form and update the membership status.	Nuraisyah binti Mohd Zikre	Harini A/P Sangaran

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Senarai Permohonan Berhenti Ahli' button.	-	Display list of pending membership termination forms.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
2	Click on the 'Dokumen Permohonan Berhenti Keahlian' hyperlink.	-	Display applicant's membership termination form.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
3	Click on the drop down button in the 'Status Permohonan' column.	-	Display the current application status by default and display all the possible statuses when clicked on the drop down button.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
4	Click on the 'Kemas Kini' button.	-	Update applicant's application status.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-

Table 15: TC002_07: Test <Approve Membership Termination Form (SD012)>

DESCRIPTION	VERSION	TEST DATE
View Monthly Report	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the monthly report displays the correct data for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications in a graphical format for a selected month.	Damiya Aina binti Basir Abd Shammad	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Laporan Koperasi' button.	-	Display overall report for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-
2	Choose a specific month and year from the dropdown or calendar.	Selected month & year: January 2024	Graph diisplays report for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications for the selected month.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-

Table 16: TC002_08: Test <View Monthly Report (SD013)>

DESCRIPTION	VERSION	TEST DATE
View Annual Report	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the annual report displays the correct data for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications in a graphical format for a selected year.	Harini A/P Sangaran	Damiya Aina binti Basir Abd Shammad

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Laporan Koperasi' button.	-	Display overall report for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications.	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-
2	Choose a specific year from the dropdown or calendar.	Selected year: 2024	Graph displays report for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications for the selected year.	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-

Table 17: TC002_09: Test <View Annual Report (SD014)>

DESCRIPTION	VERSION	TEST DATE
Add New Admin	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admin can add a new admin to the system.	Nuraisyah binti Mohd Zikre	Harini A/P Sangaran

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Tambah Admin' button.	-	Display a column to enter an identity card number for the new admin.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
2	Enter the Identity Card (IC) number of the new admin and click on 'Tambah Admin' button.	IC number : 04120107 0330	Displays appropriate message for successful admission of the new admin.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-

Table 18: TC002_10: Test <Add New Admin (SD015)>

DESCRIPTION	VERSION	TEST DATE
Add New ALK	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admin can add a new ALK to the system.	Nurul Ika Syafiny binti Azhar	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Tambah ALK' button.	-	Display a column to enter an identity card number for the new ALK.	As Expected	Pass	Harini A/P Sangaran	-
2	Enter the Identity Card (IC) number of the new ALK and click on 'Tambah ALK' button.	IC number : 04120107 0330	Displays appropriate message for successful admission of the new ALK.	As Expected	Pass	Harini A/P Sangaran	-

Table 19: TC002_11: Test <Add New ALK (SD016)>

TC003: Test ALK Subsystem: <User Story 3 (US003)>

This test contains the following test cases:

- (a) TC003_01: Test <View ALK Menu (SD026)>
- (b) TC003_02: Test <View Validated Member Application Form (SD017)>
- (c) TC003_03: Test <View Validated Loan Application Form (SD018)>
- (d) TC003_04: Test <Approve Member Application (SD019)>
- (e) TC003_05: Test <Approve Loan Application (SD020)>
- (f) TC002_06: Test <View Monthly Report (SD021)>
- (g) TC002_07: Test <View Annual Report (SD022)>

DESCRIPTION	VERSION	TEST DATE
View ALK Menu	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the ALKs are able to view the appropriate homepage once they have successfully logged in.	Nurul Ika Syafiny binti Azhar	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	ALK ID: 0001 Password : 1234	Credential can be entered.	As Expected	Pass	Harini A/P Sangaran	-
2	View ALK homepage.	-	Display the features for ALK to do in the system.	As Expected	Pass	Harini A/P Sangaran	-

Table 20: TC003_01: Test <View ALK Menu (SD026)>

DESCRIPTION	VERSION	TEST DATE
View Validated Member Application Form	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the ALKs are able to view the list of reviewed member application forms.	Harini A/P Sangaran	Damiya Aina binti Basir Abd Shammad

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	View ALK homepage.	-	Display the features for ALK to do in the system.	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-
2	Click on the 'Pengesahan Permohonan Ahli' button.	-	Display list of reviewed member application forms.	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-

Table 21: TC003_02: Test <View Validated Member Application Form (SD017)>

DESCRIPTION	VERSION	TEST DATE
View Validated Loan Application Form	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the ALKs are able to view the list of reviewed loan application forms.	Damiya Aina binti Basir Abd Shammad	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	View ALK homepage.	-	Display the features for ALK to do in the system.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-
2	Click on the 'Pengesahani Permohonan Pinjaman' button.	-	Display list of reviewed loan application forms.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-

Table 22: TC003_03: Test <View Validated Loan Application Form (SD018)>

DESCRIPTION	VERSION	TEST DATE
Approve Member Application	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the ALKs are able to view applicant's verification documents and update the application status.	Nuraisyah binti Mohd Zikre	Harini A/P Sangaran

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Pengesahan Permohonan Ahli' button.	-	Display list of pending member application forms.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
2	Click on the 'Dokumen Permohonan Ahli' hyperlink.	-	Display applicant's application form.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
3	Click on the drop down button in the 'Status Permohonan' column.	-	Display the current application status by default and display all the possible	As Expected	Pass	Damiya Aina binti	-

			statuses when clicked on the drop down button.			Basir Abd Shammad	
4	Click on the 'Kemas Kini' button.	-	Update applicant's application status.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-

Table 23: TC003_04: Test <Approve Member Application (SD019)>

DESCRIPTION	VERSION	TEST DATE
Approve Loan Application	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the ALKs are able to view member's verification documents and update the loan application status.	Nurul Ika Syafiny binti Azhar	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Pengesahan Permohonan Pinjaman' button.	-	Display list of reviewed loan application forms.	As Expected	Pass	Harini A/P Sangaran	-
2	Click on the 'Dokumen Permohonan Pinjaman' hyperlink.	-	Display member's loan application form.	As Expected	Pass	Harini A/P Sangaran	-
3	Click on the 'Muat Turun Dokumen' button.	-	Download member's verification documents for the loan application	As Expected	Pass	Harini A/P Sangaran	-
4	Click on the drop down button in the	-	Display the current application status by default and display	As Expected	Pass	Harini A/P Sangaran	-

	'Status Permohonan' column.		all the possible statuses when clicked on the drop down button.				
5	Click on the 'Kemas Kini' button.	-	Update member's loan application status.	As Expected	Pass	Harini A/P Sangaran	-

Table 24: TC003_05: Test <Approve Loan Application (SD020)>

DESCRIPTION	VERSION	TEST DATE
View Monthly Report	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the monthly report displays the correct data for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications in a graphical format for a selected month.	Damiya Aina binti Basir Abd Shammad	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Laporan Tahunan' button.	-	Display overall report for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-
2	Choose a specific month and year from the dropdown or calendar.	Selected month & year: January 2024	Graph displays report for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications for the selected month.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-

Table 25: TC003_06: Test <View Monthly Report (SD021)>

DESCRIPTION	VERSION	TEST DATE
View Annual Report	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the annual report displays the correct data for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications in a graphical format for a selected year.	Harini A/P Sangaran	Damiya Aina binti Basir Abd Shammad

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Laporan Tahunan' button.	-	Display overall report for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-
2	Choose a specific year from the dropdown or calendar.	Selected year: 2024	Graph displays report for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications for the selected year.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-

Table 26: TC003_07: Test <View Annual Report (SD034)>

TC004: Test Non-Member Subsystem: <User Story 4 (US004)>

This test contains the following test cases:

- (a) TC004_01: Test <Apply Membership (SD023)>
- (b) TC004_02: Test <View Homepage (SD027)>
- (c) TC004_03: Test <Get Member Approval (SD028)>

DESCRIPTION	VERSION	TEST DATE
Apply Membership	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
allows non-members to apply for membership by providing all necessary information and documents and that the application is successfully recorded in the system for staff and ALK review.	Nurul Ika Syafiny binti Azhar	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Register into the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
2	Log in back to the Koperasi KADA system with a valid id and password	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	
3	Click on the “Daftar Sekarang” to apply for the membership	-	Successfully load the membership form section, displaying all the membership application form	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	
4	Fill in the membership	nama: Ali bin Abu	Form is completed	As Expected	Pass	Damiya Aina binti	

	application form fields (e.g: nama, no. k/p, etc)	no.k/p: 76041303 7761				Basir Abd Shammad	
5	Click the "Simpan Maklumat" button after completing the form.	-	successfully submit the membership application form	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	

Table 27: TC004_01: Test <Apply Membership (SD023)>

DESCRIPTION	VERSION	TEST DATE
View Homepage	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
Ensure that the homepage is accessible to all users before registration or login and displays relevant information along with clear options for registration and login without exposing any member-specific details.	Harini A/P Sangaran	Damiya Aina binti Basir Abd Shammad

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	View homepage	-	Display the information of the Koperasi KADA	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-

Table 28: TC004_02: Test <View Homepage (SD027)>

DESCRIPTION	VERSION	TEST DATE
Get Member Approval	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To ensure that the system correctly retrieves and displays the membership application status for a non-member. If the application is approved, the system should provide login credentials. If the application is pending, rejected, or under review, the system should display an appropriate status message.	Harini A/P Sangaran	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	View homepage	-	Display the information of the Koperasi KADA	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
2	Click on "Semak Permohonan" button	-	Membership status page is loaded	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	
3	Enter IC number into the form	IC num: 770605030402	System accepts the IC number input	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	
4	Click "Semak Akaun"	-	System processes the request	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	
5	If approved, display credentials	id no: 0001 password: 1234	ID and password are displayed	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	

6	If pending, display "Pending" message	IC num: 770605030402(pending)	"Permohonan anda sedang diproses" message is displayed	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	
7	If rejected, display "Rejected" message	IC num: 770605030402(rejected)	"Permohonan anda ditolak" message is displayed	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	
8	If in reviewed, display "in reviewed" message	IC num: 770605030402(in reviewed)	"Permohonan anda sedang dinilai" message is displayed	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	

Table 29: TC004_03: Test <Get Member Approval (SD028)

DESCRIPTION	VERSION	TEST DATE
Log in	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the users can log in to the system by entering valid identity number and password.	Nurul Ika Syafiny binti Azhar	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password for any user.	Chosen user : admin Admin ID: 0001 Password : 1234	Credentials can be entered.	As Expected	Pass	Harini A/P Sangaran	-
2	View admin homepage.	-	Display the features for admin to do in the system	As Expected	Pass	Harini A/P Sangaran	-

Table 29: TC005_01: Test <Log in (SD029)>

DESCRIPTION	VERSION	TEST DATE
Log out	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the users can log in to the system by entering valid identity number and password.	Harini A/P Sangaran	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	View homepage.	-	Display the features for user to do in the system.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-
2	Click on 'Daftar Keluar' button.	-	Log out of the system and display general homepage.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-

Table 30: TC005_02: Test <Log out (SD030)>

TC005: Test Loan Subsystem: <User Story 6 (US006)>

This test contains the following test cases:

- (a) TC006_01: Test <Apply Loan (SD001)>
- (b) TC006_02: Test <Check Loan Approval (SD002)>
- (c) TC006_03: Test <Find Transaction History (SD004)>
- (d) TC006_04: Test <Check Loan Balance (SD005)>

DESCRIPTION	VERSION	TEST DATE
Apply Loan	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
to verify that a member can successfully apply for a loan after logging in	Damiya Aina binti Basir Abd Shammad	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Harini A/P Sangaran	-
2	Navigate to the Mohon Pinjaman section from the Member homepage.	-	Successfully load the loan form section, displaying all the loan application form	As Expected	Pass	Harini A/P Sangaran	
3	Fill in the loan application form fields (e.g: tempoh bayaran balik, jumlah pinjaman, etc)	Tempoh Bayaran Balik: 36 bulan Jumlah Pinjaman: RM10000	Form is complete	As Expected	Pass	Harini A/P Sangaran	

4	Upload all required documents (e.g., salary slips, IC copies, etc).	Salary Slip: uploaded IC copies: uploaded	Successfully upload the file required and can submit the form	As Expected	Pass	Harini A/P Sangaran	
5	Click the "Submit" button after completing the form.	-	successfully submit the loan application form	As Expected	Pass	Harini A/P Sangaran	

Table 31: TC006_01: Test <Apply Loan (SD001)>

DESCRIPTION	VERSION	TEST DATE
Check Loan Approval	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
to verify that loan application status is displayed correctly	Harini A/P Sangaran	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Nuraisyah binti Mohd Zikre	-
2	Navigate to the Semak Kelulusan Pinjaman section from the Member homepage.	-	Successfully load the loan application status section, displaying all the information of the member with the loan status	As Expected	Pass	Nuraisyah binti Mohd Zikre	
3	Check the status of loan apply	-	Loan Status is appeared successfully	As Expected	Pass	Nuraisyah binti Mohd Zikre	

Table 32: TC006_02: Test <Check Loan Approval (SD002)>

DESCRIPTION	VERSION	TEST DATE
Find Transaction History	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
Ensure that members can access a detailed, accurate transaction history with relevant filters to provide clarity on past transactions	Harini A/P Sangaran	Damiya Aina binti Basir Abd Shammad

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-
2	Navigate to the Semak Penyata section from the Member homepage.	-	Successfully load the member invoice section, displaying all the information of the member financial history	As Expected	Pass	Nurul Ika Syafiny binti Azhar	
3	view the table of member's finance	-	table of member's finance is successfully appear	As Expected	Pass	Nurul Ika Syafiny binti Azhar	

Table 33: TC006_03: Test <Find Transaction History (SD004)>

DESCRIPTION	VERSION	TEST DATE
Check Loan Balance	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
Ensure that the system accurately displays the member's current loan balance, repayment history, and any outstanding amounts to help members keep track of their loan obligations.	Damiya Aina binti Basir Abd Shamad	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Harini A/P Sangaran	-
2	Navigate to the Baki Pinjaman section from the Member homepage.	-	Successfully load the loan balance section, displaying the loan balance amount and also the repayment history	As Expected	Pass	Harini A/P Sangaran	
3	view the table of loan repayment history and also the loan balance amount	-	table of repayment history and loan balance amount is successfully appear	As Expected	Pass	Harini A/P Sangaranr	

Table 34: TC006_04: Test <Check Loan Balance (SD005)>

11.2 Beta Testing

11.2.0 Methods Class

Entity Name	NonMember
Method Name	applyForMembership
Input	Non-member details (name, ID number, email, marital status, gender, religion, ethnicity, home address, postcode, state, membership number, office details, phone contacts, salary, and other financial details)
Output	Membership application submitted and stored in the database
Algorithm	1. Start 2. Non-member accesses the membership application form page via website. 3. Non-member fills in the required information 4. System validates input fields 5. If the information is valid, the form is submitted 6. Else, an error message is displayed, asking for corrections 7. System stores applications in the database with 'Pending' status 8. Admin reviews the application and either approves or rejects it 9. If approved, applicants can check for their login details via the website. 10. If rejected, the system notifies the applicant when they check their application via the website 11. End

Table 35: Method Class of Beta Testing: Membership Application

11.2.1 Flow Graph

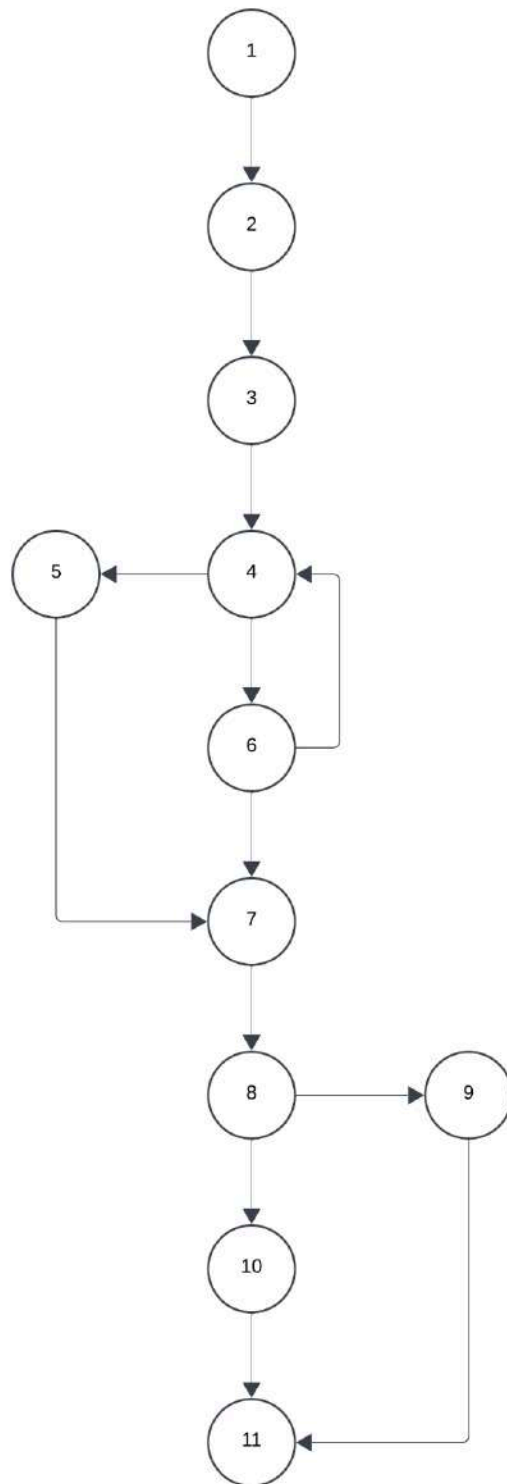


Figure 77: Flow Chart of Beta Testing: Membership Application

11.2.2 Cyclomatic Complexity Calculation

Using different formulas:

1. Formula 1: $V(G) = E - N + 2$
 - Edges (E) = 13, Nodes (N) = 11
 - $V(G) = 13 - 11 + 2 = 3$
2. Formula 2: $V(G) = P + 1$
 - Decision points (P) = 2
 - $V(G) = 2 + 1 = 3$
3. Formula 3: Number of regions in the flow graph
 - $V(G) = 3$

Thus, the Cyclomatic Complexity is 3, meaning there are three independent paths in the program.

11.2.3 Test Cases

Test Case No.	Independent Path	Test Data	Expected Result
TC001	1 → 2 → 3 → 4 → 5 → 7 → 8 → 9 → 11	Non-member fills the form correctly and submits	Membership application stored in the database and approved
TC002	1 → 2 → 3 → 4 → 6 → 4 (loop) → 5 → 7 → 8 → 10 → 11	Non-member enters invalid input and submits	System rejects form and prompts for corrections before resubmission
TC003	1 → 2 → 3 → 4 → 5 → 7 → 8 → 10 → 11	Form submission disapproved by ALK or Admin due to missing fields or invalid information.	System displays fail application when applicator check their applications

Table 36: Test Cases of Beta Testing: Membership Application

12.0 UAT Reports

Date: 23/1/2025

Tested by: En. Ahmad Rohailan bin Hani, representative of Koperasi KADA Sdn. Bhd.

Feedback:

- System process is accurate to the current existing business process.
- Need to add Member Termination Application as well into the system.
- Include the report for the quantity of inactive and active members throughout the month.

13.0 Gantt Chart

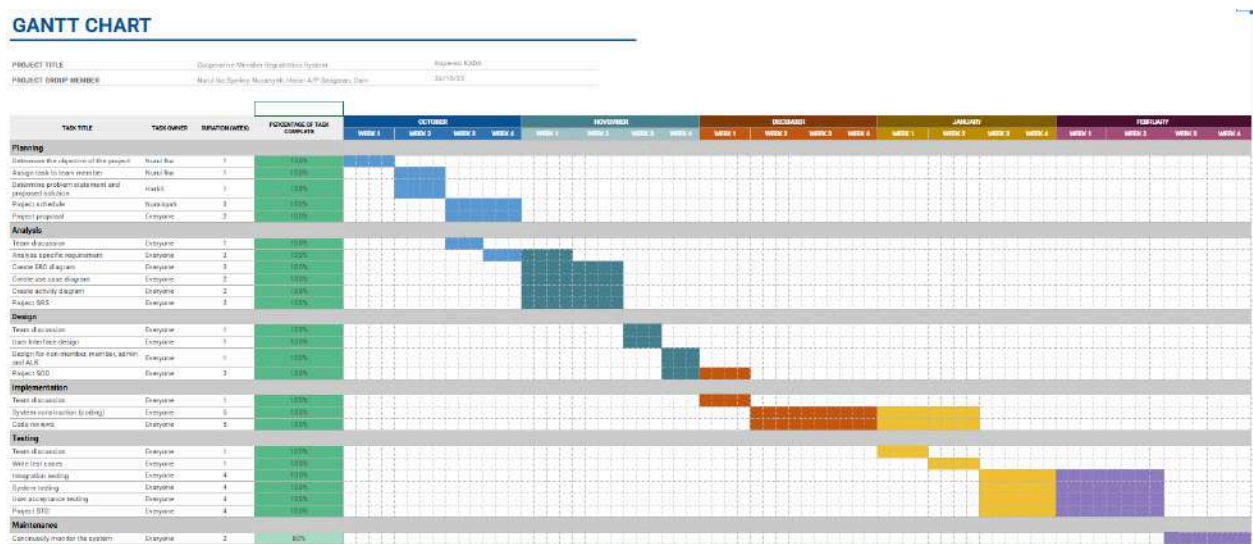


Figure 78: Gantt Chart

Reference Link:

https://docs.google.com/spreadsheets/d/1MncYaFx7WbLbXWMqy9_3xP6GLY3oKiqbybpXH_Mi-r4/edit?usp=sharing